

Davenport Perspective

William Klugh

Contents

Introduction	3
Exploring	4
Univariate Exploring.....	4
m_mcd_payermix_patientDays_lg – Log of Medicaid patient days / total patient days .	4
m_mcr_payermix_patientDays – Medicare patient days / total patient days	5
f_gross_ip_rev_adj_lg – Gross Inpatient Revenue log adjusted for patient days	6
f_gross_op_rev_adj_lg – Gross Outpatient Revenue log adjusted for patient days	7
f_net_inc_adj – Net Income (Profit) adjusted for patient days	8
f_net_patient_rev_adj_lg – Net Patient Revenue log adjusted for patient days	9
f_nonpatient_rev_adj_lg – Total Non Patient Revenue log adjusted for patient days ...	10
qm_mort_30_pn – Death rate for pneumonia patients	11
qm_mort_30_stk – Death rate for stroke patients	11
qo_pc_01_lg – Logged Percent of mothers whose deliveries were scheduled too early, when a scheduled delivery was not medically necessary (1-2 weeks early)	12
qo_sep_1 – Severe Sepsis and Septic Shock	13
qr_edac_30_ami – Hospital return days for heart attack patients	13
qr_readm_30_pn – Rate of readmission for pneumonia patients	14
qt_op_18b – Median time patients spent in the emergency department before leaving from the visit.....	14
qt_edv – Emergency department volume.....	15
Bivariate Exploring.....	16
Testing Section	41
Confirming Section	46
Conclusions	47

Introduction

I don't have a specific current event in mind, but Medicare and Medicaid programs seem to be hot topics currently which has them on my mind. I'd like to learn about the relationship between Medicare/Medicaid programs and the financial performance of hospitals. I would also like to learn more about the relationship between the quality and financial performance of hospitals. I think that quality and Medicare/Medicaid programs likely do impact the financial performance of hospitals.

Question 1:

What is the relationship between quality and financial performance?

Question 2

What is the relationship between financial performance and the Medicare and Medicaid programs?

Medicare and Medicaid Variables:

- m_mcd_payermix_patientDays_lg – Medicaid patient days / total patient days log
- m_mcr_payermix_patientDays – Medicare patient days / total patient days

Financial Variables:

- f_gross_ip_rev_adj_lg – Gross Inpatient Revenue log adjusted for patient days
- f_gross_op_rev_adj_lg – Gross Outpatient Revenue log adjusted for patient days
- f_net_inc_adj – Net Income (Profit) adjusted for patient days
- f_net_patient_rev_adj_lg – Net Patient Revenue log adjusted for patient days
- f_nonpatient_rev_adj_lg – Total Non Patient Revenue log adjusted for patient days

Quality Variables:

- qm_mort_30_pn – Death rate for pneumonia patients
- qm_mort_30_stk – Death rate for stroke patients
- qo_pc_01_lg – Logged Percent of mothers whose deliveries were scheduled too early, when a scheduled delivery was not medically necessary (1-2 weeks early)
- qo_sep_1 – Severe Sepsis and Septic Shock
- qr_edac_30_ami – Hospital return days for heart attack patients
- qr_readm_30_pn – Rate of readmission for pneumonia patients
- qt_op_18b – Median time patients spent in the emergency department before leaving from the visit
- qt_edv – Emergency department volume

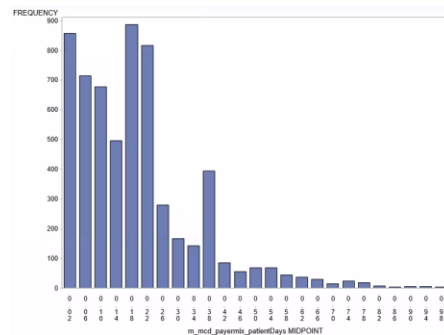
Exploring

Univariate Exploring

Medicare and Medicaid Variables:

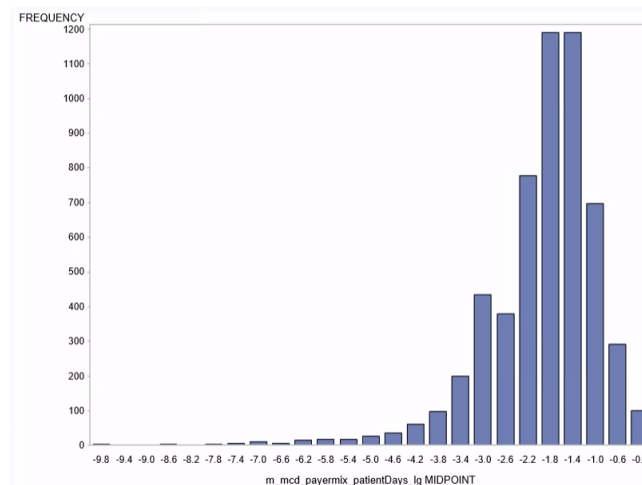
m_mcd_payermix_patientDays_lg – Log of Medicaid patient days / total patient days

Before Taking Log



With all data

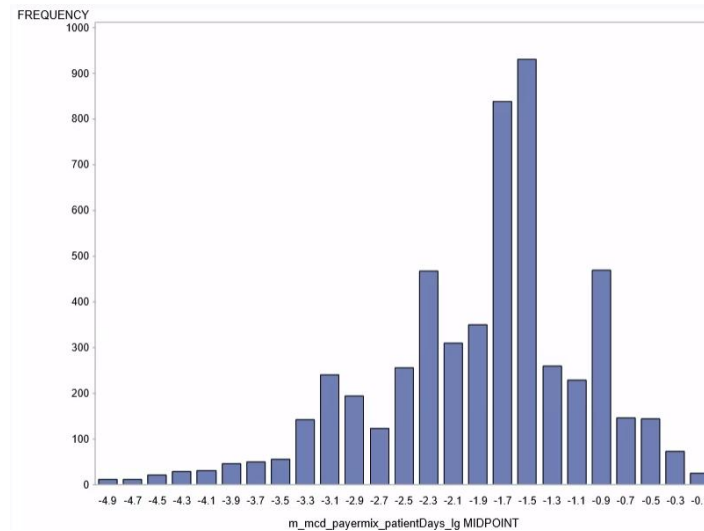
Analysis Variable : m_mcd_payermix_patientDays_lg									
Mean	Std Dev	Minimum	Maximum	N	1st Pctl	5th Pctl	Median	95th Pctl	99th Pctl
-1.9386512	0.9892241	-9.8378818	-0.0065897	5551	-5.4662019	-3.6529775	-1.7174510	-0.6551864	-0.2836661



The Medicaid payer mix log shows substantial outliers that will need to be removed in order to use standard regression tools.

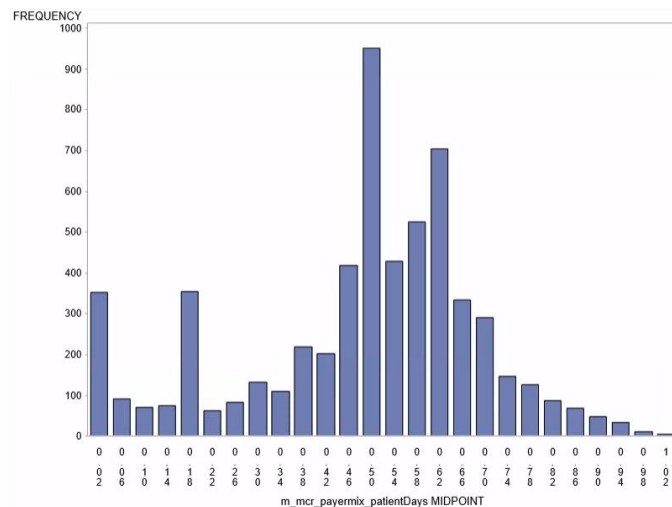
With outliers removed (m_mcd_payermix_patientDays_lg between -5 and 0)

Analysis Variable : m_mcd_payermix_patientDays_lg									
Mean	Std Dev	Minimum	Maximum	N	1st Pctl	5th Pctl	Median	95th Pctl	99th Pctl
-1.8750100	0.8468476	-4.9921318	-0.0065897	5466	-4.3578736	-3.3816528	-1.7151571	-0.6470548	-0.2829555



m_mcr_payermix_patientDays – Medicare patient days / total patient days

Analysis Variable : m_mcr_payermix_patientDays									
Mean	Std Dev	Minimum	Maximum	N	1st Pctl	5th Pctl	Median	95th Pctl	99th Pctl
0.4831290	0.2133453	0	1.0349265	5925	0	0.0078111	0.5158630	0.7881557	0.9122596



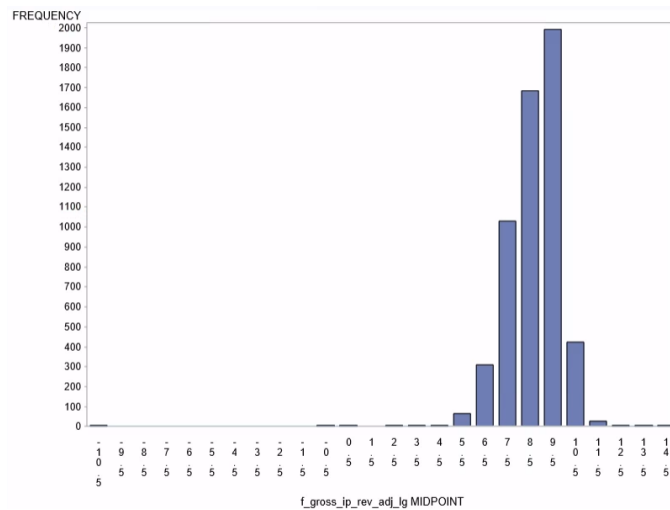
The Medicare payer mix has a somewhat normal shape and no substantial outliers.

Financial Variables:

f_gross_ip_rev_adj_lg – Gross Inpatient Revenue log adjusted for patient days

With all data

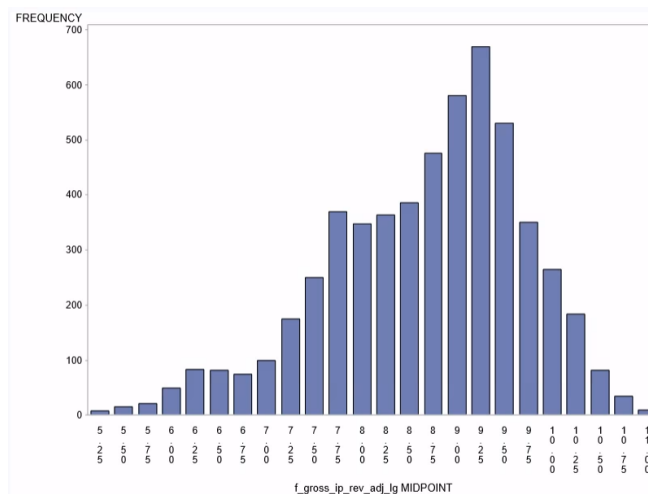
Analysis Variable : f_gross_ip_rev_adj_lg									
Mean	Std Dev	Minimum	Maximum	N	1st Pctl	5th Pctl	Median	95th Pctl	99th Pctl
8.6796391	1.1164965	-10.3401283	14.5616730	5548	5.8744449	6.6550567	8.8568521	10.1918551	10.7552055



Gross inpatient Revenue adjusted for patient days and logged shows substantial outliers. These will need to be trimmed in order to use our standard regression tools.

With outliers removed (f_gross_ip_rev_adj_lg between 5 and 11)

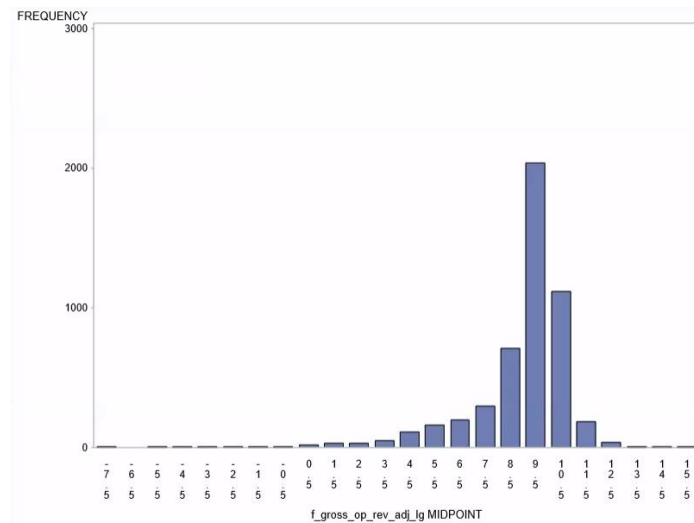
Analysis Variable : f_gross_ip_rev_adj_lg									
Mean	Std Dev	Minimum	Maximum	N	1st Pctl	5th Pctl	Median	95th Pctl	99th Pctl
8.6762340	1.0338620	5.1635452	10.9871017	5504	5.9611013	6.6912364	8.8532820	10.1593540	10.5769971



f_gross_op_rev_adj_lg – Gross Outpatient Revenue log adjusted for patient days

With all data

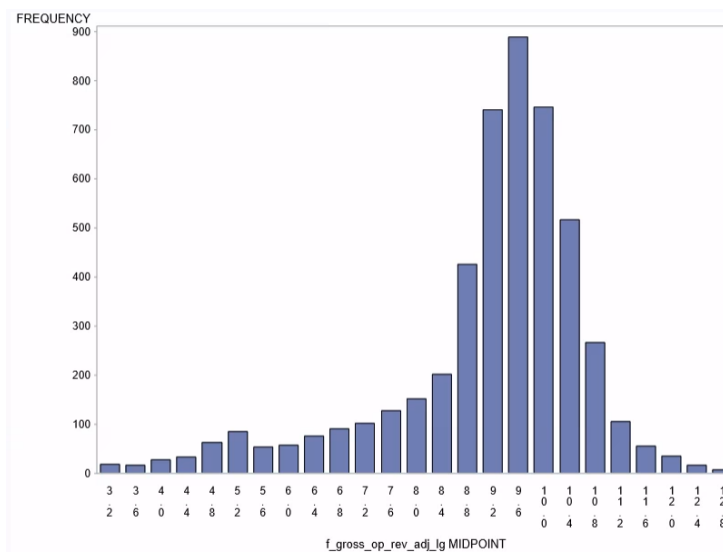
Analysis Variable : f_gross_op_rev_adj_lg									
Mean	Std Dev	Minimum	Maximum	N	1st Pctl	5th Pctl	Median	95th Pctl	99th Pctl
8.9274314	2.1025960	-7.8326092	15.6575717	5060	0.6672250	4.8006712	9.4547667	10.9563572	12.0176808



Gross Outpatient Revenue log adjusted for patient days shows substantial outliers that will need to be removed in order to use our standard regression tools.

With outliers removed (f_gross_op_rev_adj_lg between 3 and 13)

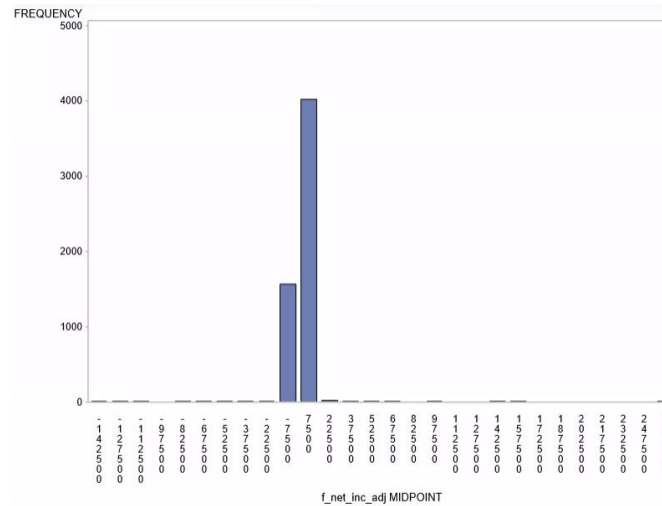
Analysis Variable : f_gross_op_rev_adj_lg									
Mean	Std Dev	Minimum	Maximum	N	1st Pctl	5th Pctl	Median	95th Pctl	99th Pctl
9.1291762	1.5764876	3.0600678	12.9657854	4925	3.9901150	5.3855918	9.4765432	10.9399967	11.9125760



f_net_inc_adj – Net Income (Profit) adjusted for patient days

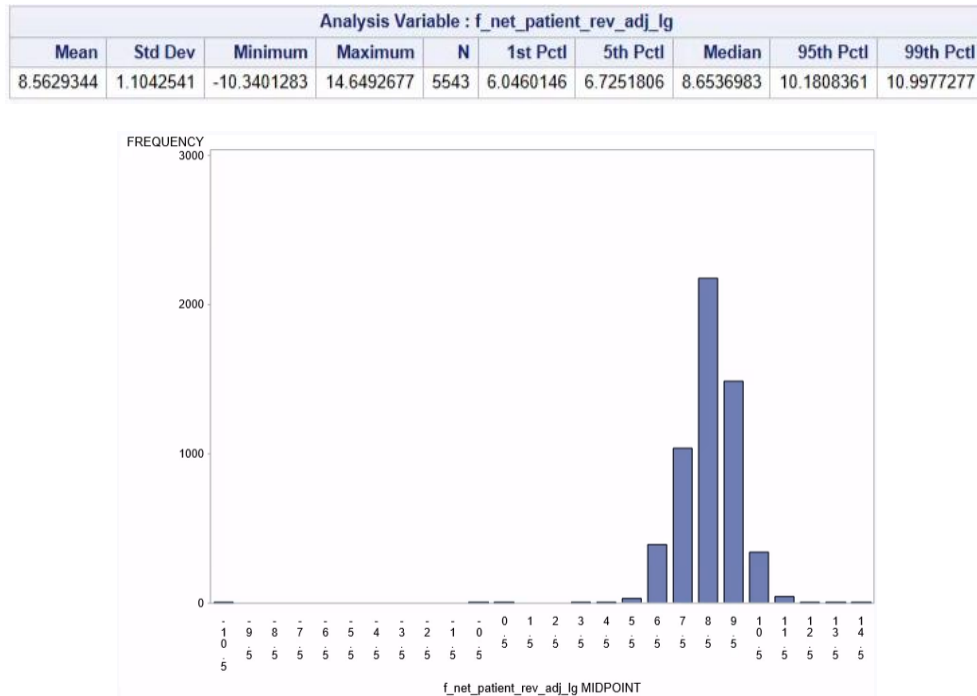
With all data

Analysis Variable : f_net_inc_adj									
Mean	Std Dev	Minimum	Maximum	N	1st Pctl	5th Pctl	Median	95th Pctl	99th Pctl
606.8564953	7945.77	-142878.83	259349.42	5686	-10336.42	-1314.21	293.3802492	4497.33	12821.65



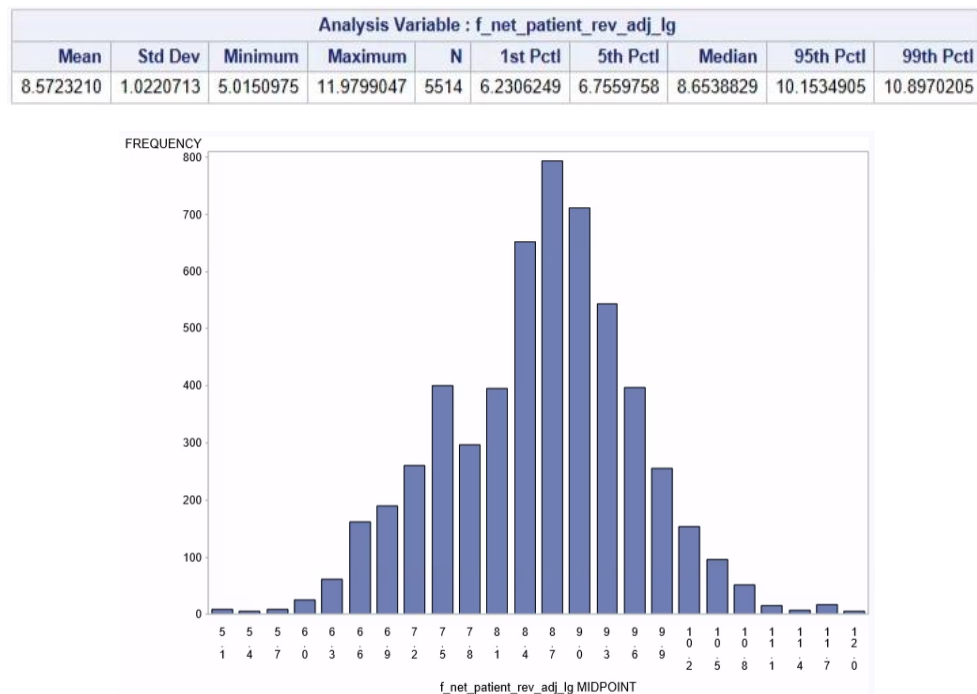
f_net_patient_rev_adj_lg – Net Patient Revenue log adjusted for patient days

With all data



Net Patient Revenue log adjusted for patient days shows substantial outliers that will need to be trimmed in order to use standard regression tools.

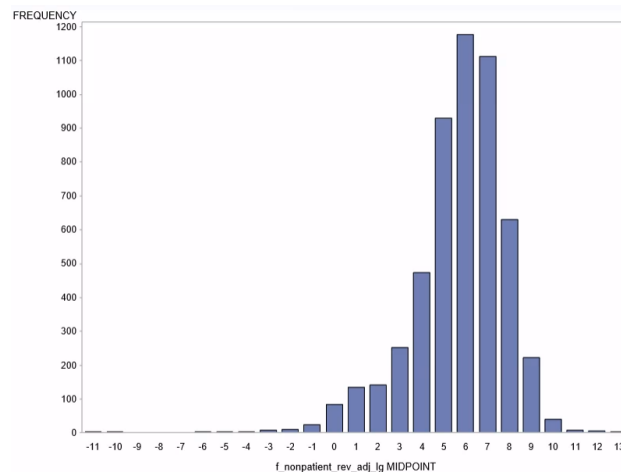
With outliers removed (f_net_patient_rev_adj_lg between 5 and 12)



f_nonpatient_rev_adj_lg – Total Non Patient Revenue log adjusted for patient days

With all data

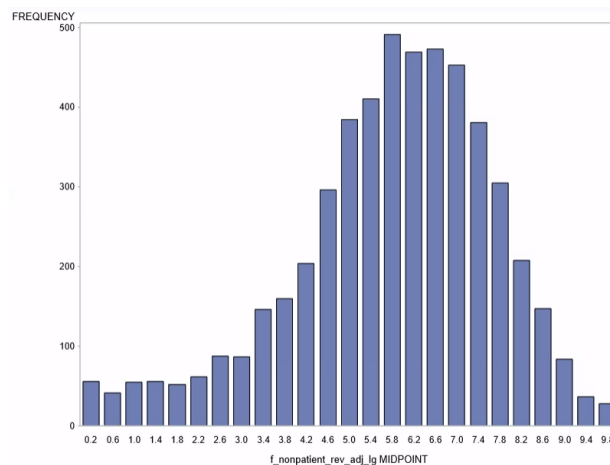
Analysis Variable : f_nonpatient_rev_adj_lg									
Mean	Std Dev	Minimum	Maximum	N	1st Pctl	5th Pctl	Median	95th Pctl	99th Pctl
5.7145287	2.0889482	-11.2084905	12.9788823	5261	-0.3481499	1.4992068	5.9787932	8.5296409	9.5416879



Total Non-Patient Revenue log adjusted for patient days shows substantial outliers. These will need to be trimmed so that standard regression tools may be used.

With outliers removed (f_nonpatient_rev_adj_lg between 0 and 10)

Analysis Variable : f_nonpatient_rev_adj_lg									
Mean	Std Dev	Minimum	Maximum	N	1st Pctl	5th Pctl	Median	95th Pctl	99th Pctl
5.7866976	1.8902121	0.0041147	9.9875344	5176	0.3666531	1.9848869	5.9979334	8.4983184	9.2900149

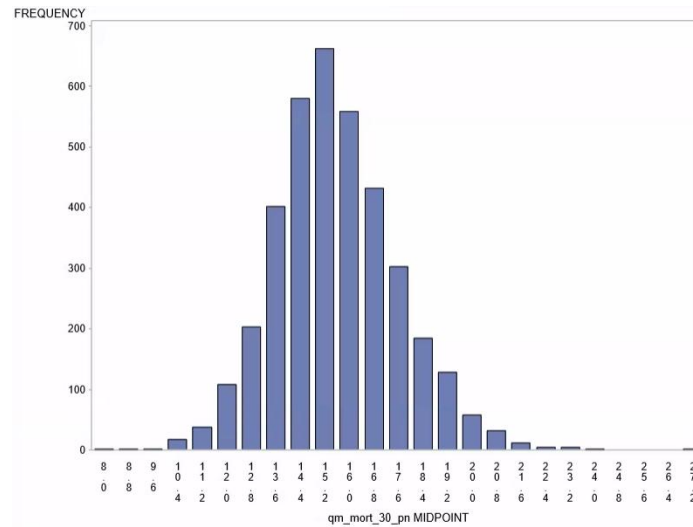


The minimum is set to zero as a total revenue of less than zero does not seem feasible. The maximum is set to ten to remove outliers.

Quality Variables:

qm_mort_30_pn – Death rate for pneumonia patients

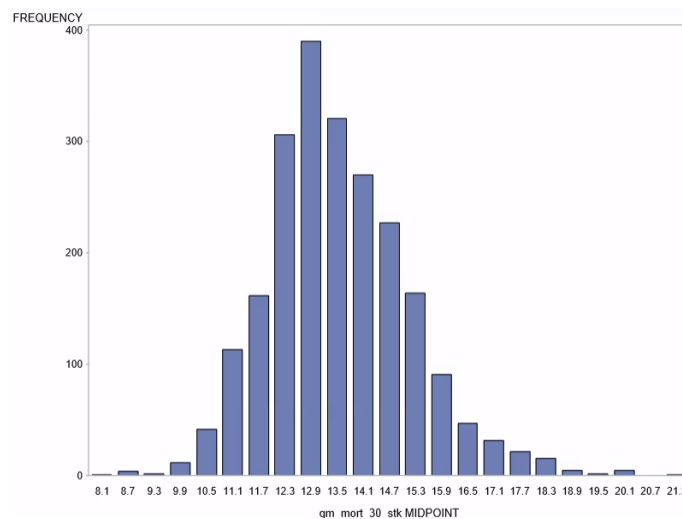
Analysis Variable : qm_mort_30_pn									
Mean	Std Dev	Minimum	Maximum	N	1st Pctl	5th Pctl	Median	95th Pctl	99th Pctl
15.5190107	2.0077889	7.7000000	27.4000000	3740	11.3000000	12.4000000	15.4000000	19.0000000	20.7000000



Death rate for pneumonia patients does not show any substantial outliers and has a somewhat normal shape.

qm_mort_30_stk – Death rate for stroke patients

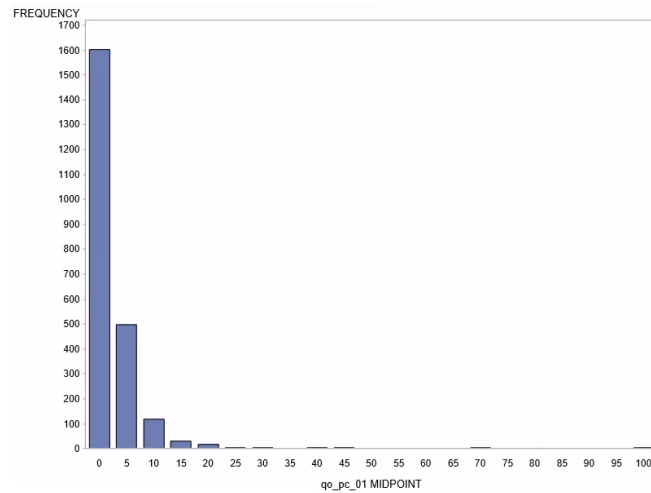
Analysis Variable : qm_mort_30_stk									
Mean	Std Dev	Minimum	Maximum	N	1st Pctl	5th Pctl	Median	95th Pctl	99th Pctl
13.4929306	1.6343899	7.9000000	21.3000000	2235	10.3000000	11.1000000	13.3000000	16.4000000	18.1000000



Death rate for stroke patients does not show any substantial outliers and has a somewhat normal shape.

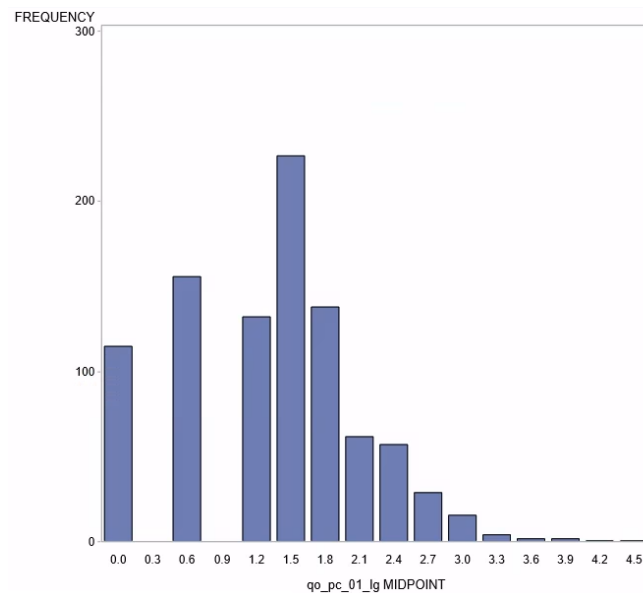
qo_pc_01_lg – Logged Percent of mothers whose deliveries were scheduled too early, when a scheduled delivery was not medically necessary (1-2 weeks early)

Before Taking Log



After Taking Log

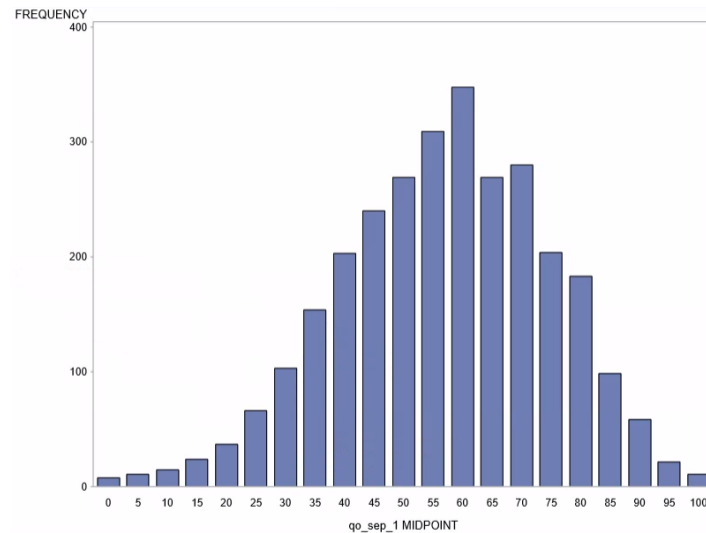
Analysis Variable : qo_pc_01_lg									
Mean	Std Dev	Minimum	Maximum	N	1st Pctl	5th Pctl	Median	95th Pctl	99th Pctl
1.3554292	0.7843790	0	4.6051702	942	0	0	1.3862944	2.6390573	3.1780538



The logged percentage of mothers whose deliveries were scheduled too early, when a scheduled delivery was not medically necessary does not show substantial outliers.

qo_sep_1 – Severe Sepsis and Septic Shock

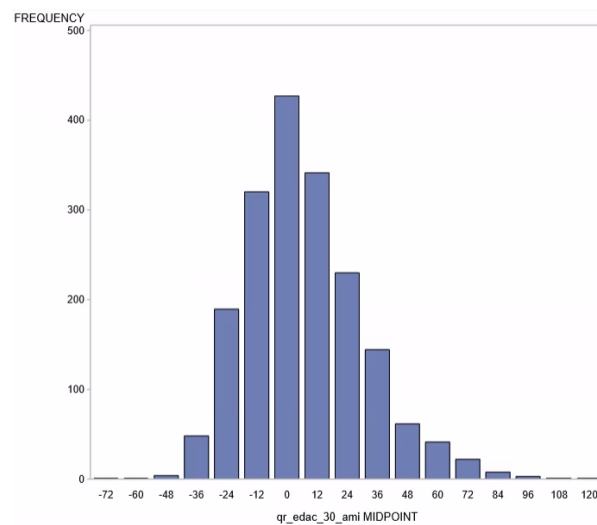
Analysis Variable : qo_sep_1									
Mean	Std Dev	Minimum	Maximum	N	1st Pctl	5th Pctl	Median	95th Pctl	99th Pctl
56.8105697	17.7798784	0	100.0000000	2914	11.0000000	26.0000000	58.0000000	84.0000000	93.0000000



Severe Sepsis and Septic Shock does not show any substantial outliers and has a seemingly normal shape.

qr_edac_30_ami – Hospital return days for heart attack patients

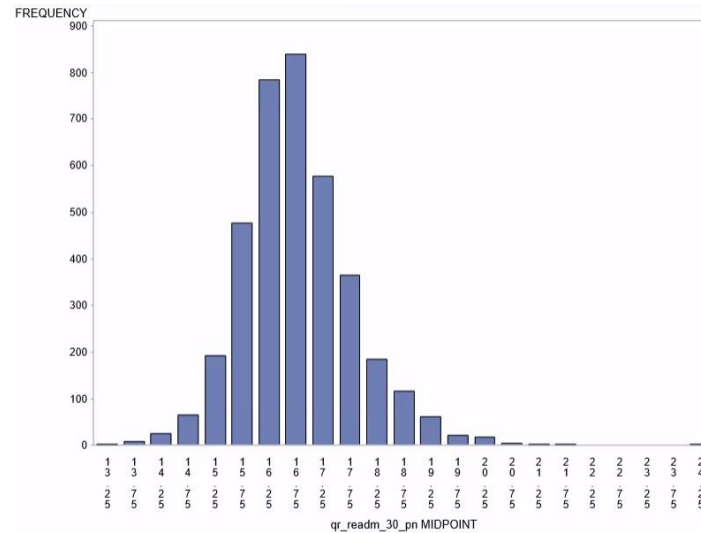
Analysis Variable : qr_edac_30_ami									
Mean	Std Dev	Minimum	Maximum	N	1st Pctl	5th Pctl	Median	95th Pctl	99th Pctl
6.7940921	23.4464156	-72.2000000	121.5000000	1845	-37.2000000	-25.7000000	4.0000000	50.0000000	76.1000000



Hospital return days for heart attack patients shows no substantial outliers, and has a somewhat normal shape.

qr_readm_30_pn – Rate of readmission for pneumonia patients

Analysis Variable : qr_readm_30_pn									
Mean	Std Dev	Minimum	Maximum	N	1st Pctl	5th Pctl	Median	95th Pctl	99th Pctl
16.7299041	1.0235829	13.2000000	24.4000000	3752	14.4000000	15.2000000	16.6000000	18.6000000	19.6000000

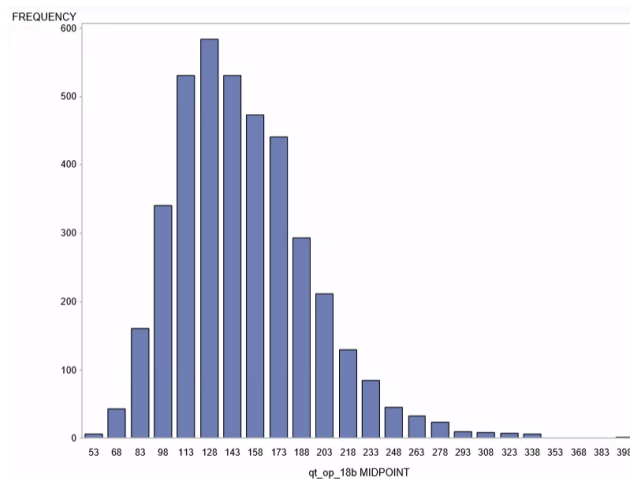


Rate of readmission for pneumonia patients shows no substantial outliers and has a somewhat normal shape.

qt_op_18b – Median time patients spent in the emergency department before leaving from the visit

With all data

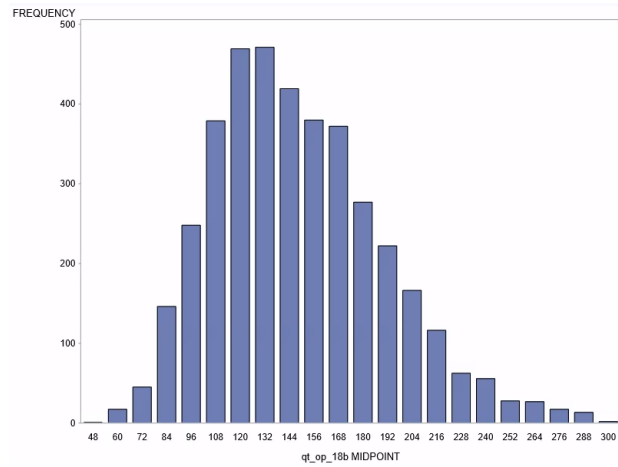
Analysis Variable : qt_op_18b									
Mean	Std Dev	Minimum	Maximum	N	1st Pctl	5th Pctl	Median	95th Pctl	99th Pctl
148.9959606	43.6379443	51.0000000	397.0000000	3961	73.0000000	88.0000000	143.0000000	228.0000000	279.0000000



Median time patients spend in the emergency department before leaving shows some substantial outliers that will need to be removed to use standard regression tools.

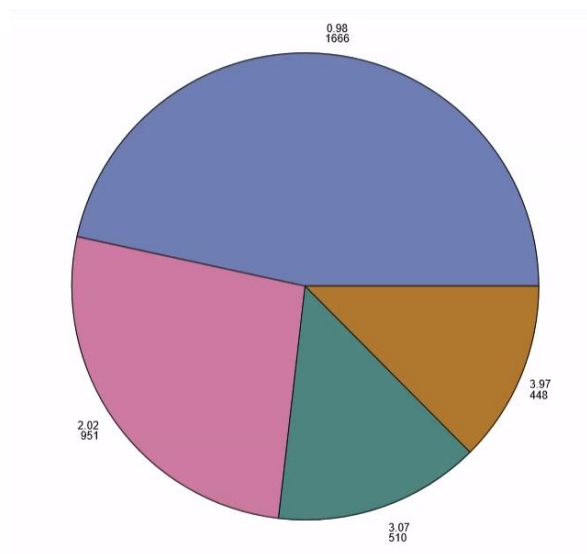
With outliers removed (qt_op_18b between 0 and 300)

Analysis Variable : qt_op_18b									
Mean	Std Dev	Minimum	Maximum	N	1st Pctl	5th Pctl	Median	95th Pctl	99th Pctl
148.0192942	41.7251051	51.0000000	300.0000000	3939	73.0000000	88.0000000	143.0000000	224.0000000	267.0000000



qt_edv – Emergency department volume

qt_edv	Frequency	Percent	Cumulative Frequency	Cumulative Percent
1	1666	46.60	1666	46.60
2	951	26.60	2617	73.20
3	510	14.27	3127	87.47
4	448	12.53	3575	100.00
Frequency Missing = 2350				

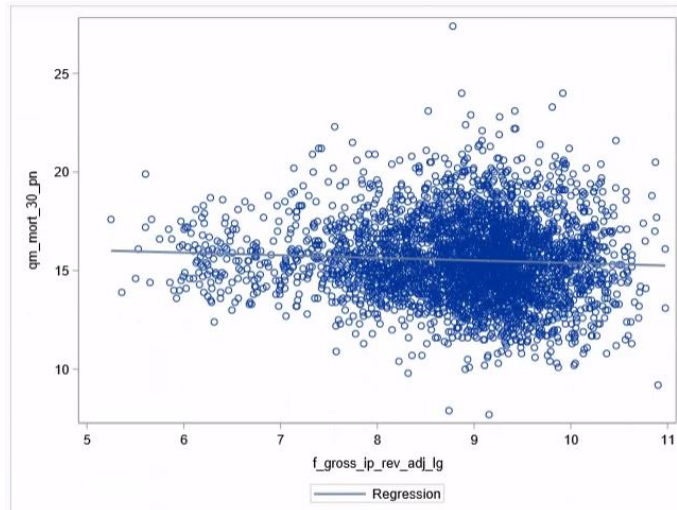


Emergency department volume is categorized into four groups with 1 being the least emergency department volume and 4 being the most.

Bivariate Exploring

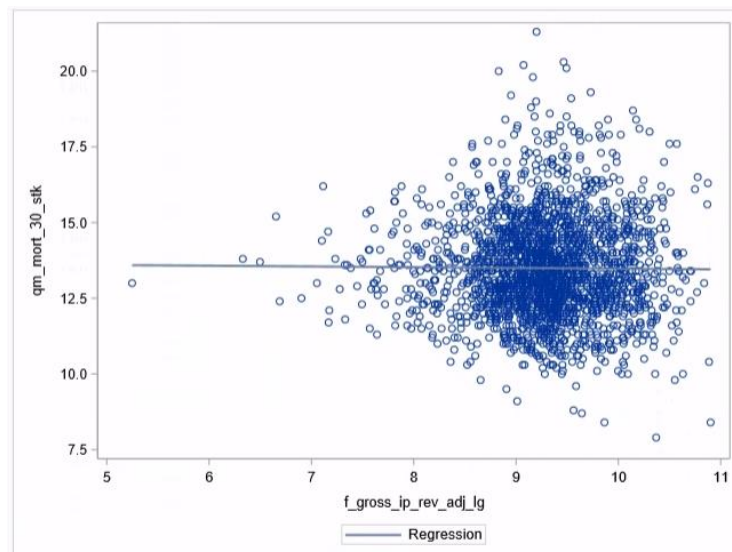
Finance vs Quality:

f_gross_ip_rev_adj_lg vs qm_mort_30_pn



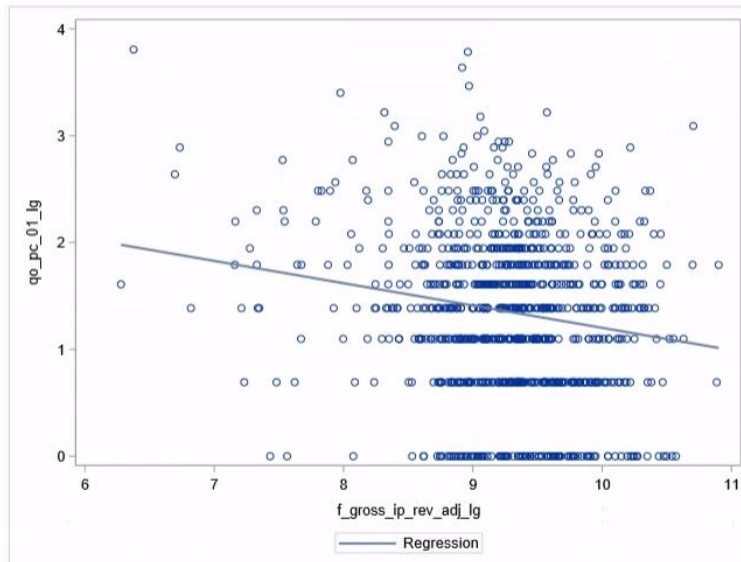
As gross inpatient revenue increases, the death rate for pneumonia patients appears to decrease. There is more variability in the death rate when gross inpatient revenue is higher.

f_gross_ip_rev_adj_lg vs qm_mort_30_stk



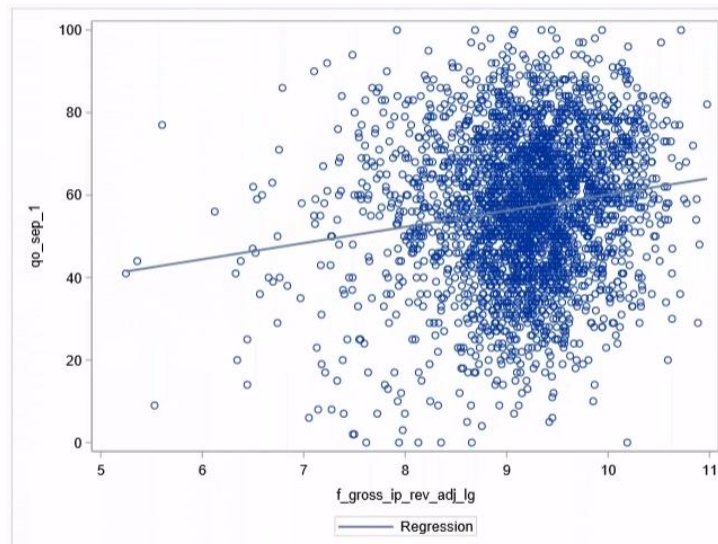
There does not appear to be much of a relationship between gross inpatient revenue and the death rate of stroke patients.

f_gross_ip_rev_adj_lg vs qo_pc_01_lg



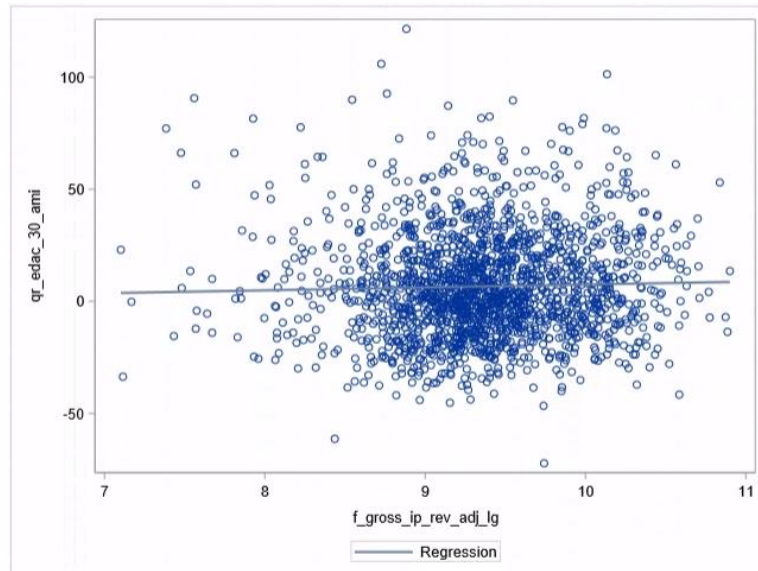
There appears to be a negative relationship between gross inpatient revenue and the percentage of mothers whose deliveries were scheduled too early.

f_gross_ip_rev_adj_lg vs qo_sep_1



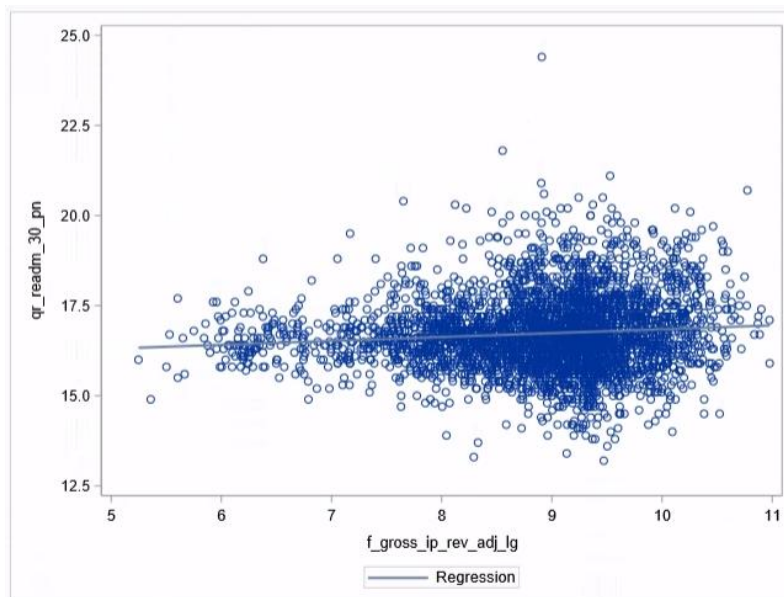
As gross inpatient revenue increases, severe sepsis and septic shock appear to increase. There is a lot of variability in severe sepsis and septic shock throughout revenue ranges.

f_gross_ip_rev_adj_lg vs qr_edac_30_ami



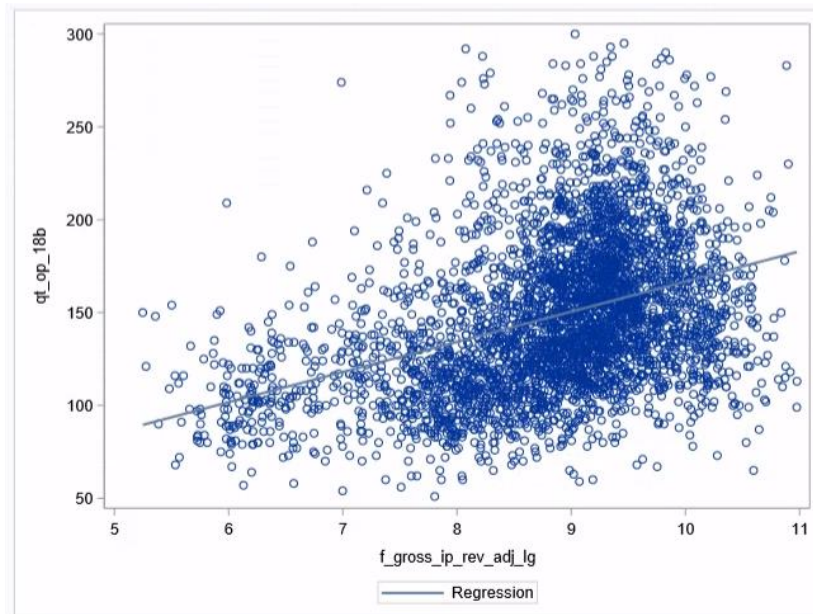
There appears to be a slight positive relationship between gross inpatient revenue and hospital return days for heart attack patients.

f_gross_ip_rev_adj_lg vs qr_readm_30_pn



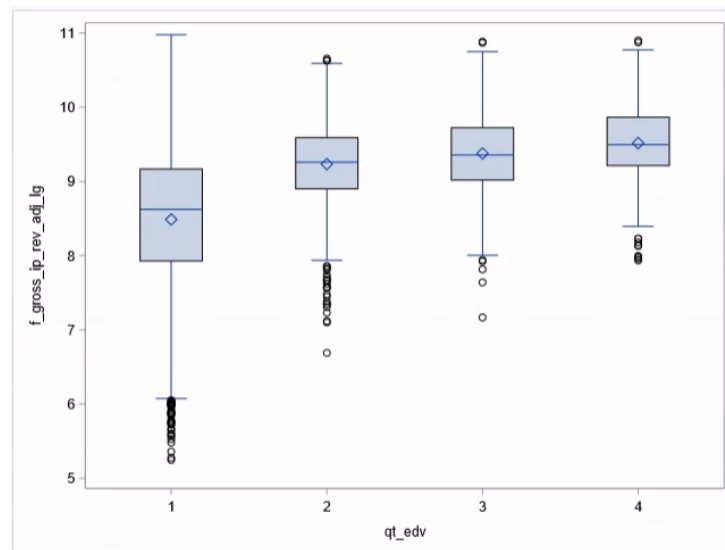
There appears to be a positive relationship between gross inpatient revenue and the readmission rate for pneumonia patients.

f_gross_ip_rev_adj_lg vs qt_op_18b



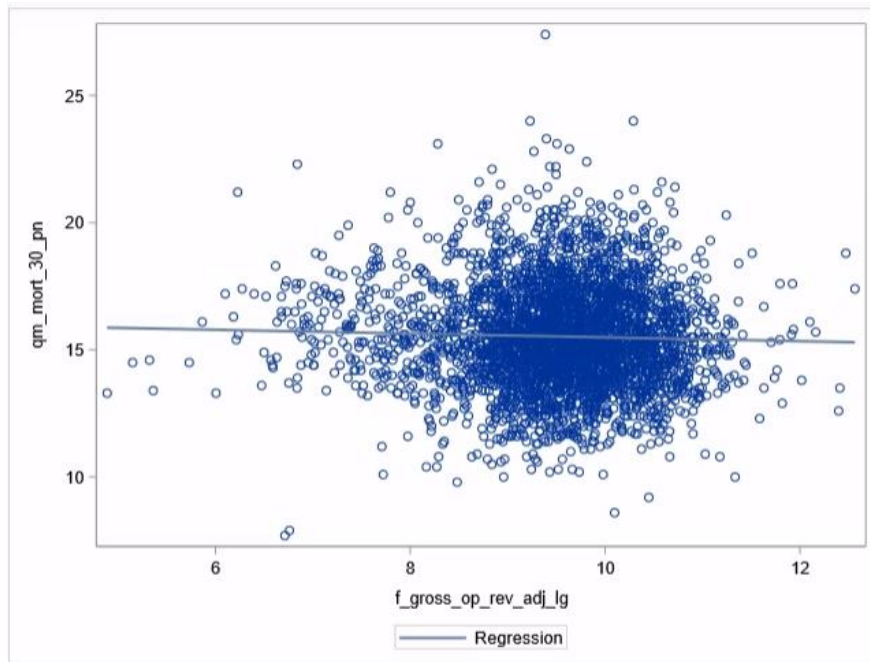
There appears to be a positive relationship between gross inpatient revenue and the median time patients spent in the emergency department before leaving from the visit. There is more variability in the median time when gross inpatient revenue is high.

f_gross_ip_rev_adj_lg vs qt_edv



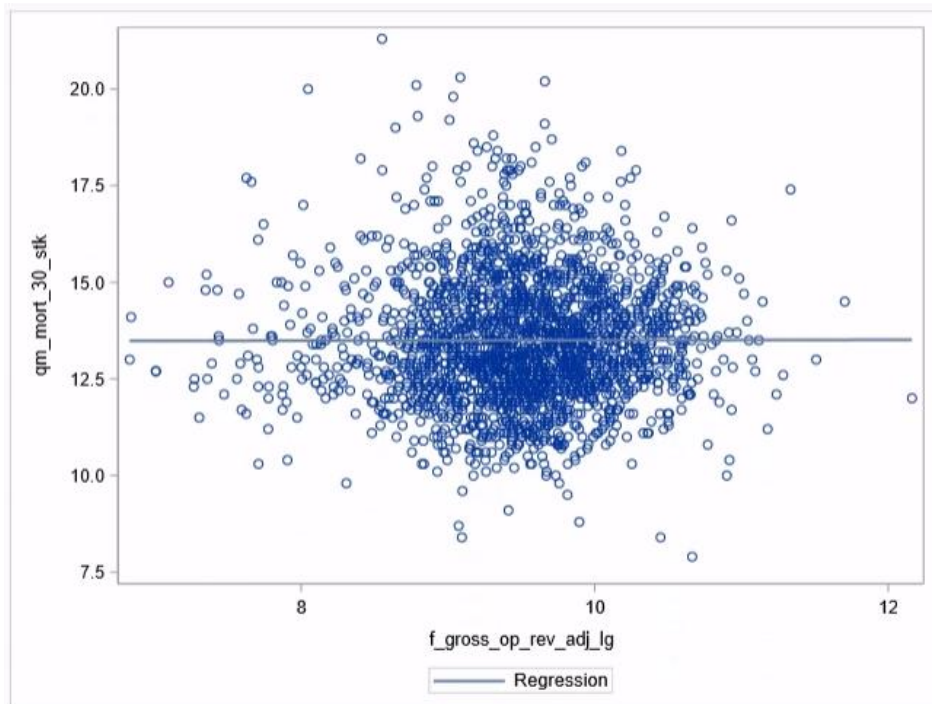
Hospitals with the lowest emergency department volume have lower gross inpatient revenue and the most variability in gross inpatient revenue.

f_gross_op_rev_adj_lg vs qm_mort_30_pn



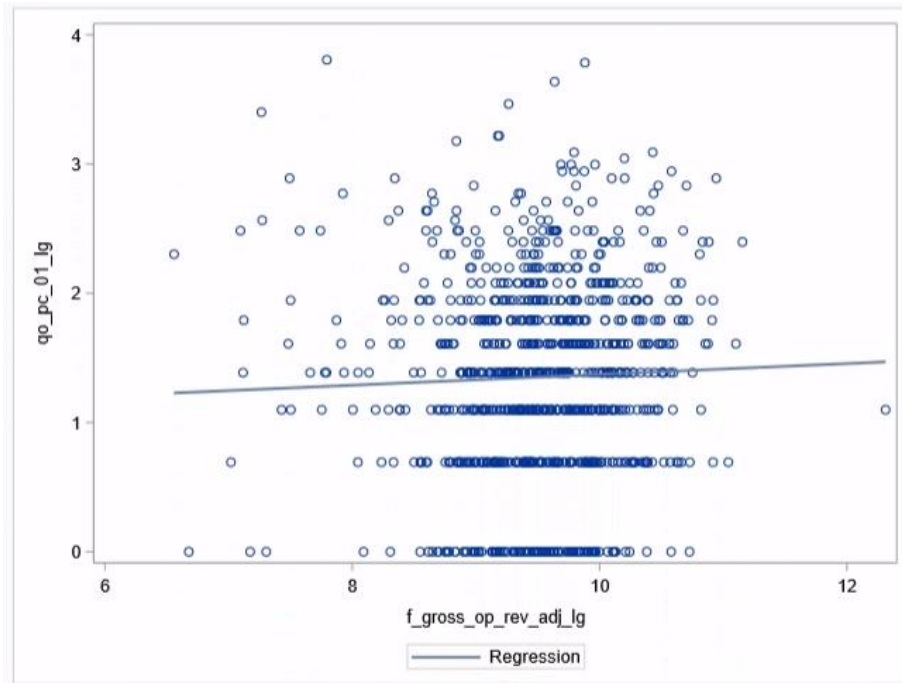
There appears to be a negative relationship between gross outpatient revenue and the death rate of pneumonia patients.

f_gross_op_rev_adj_lg vs qm_mort_30_stk



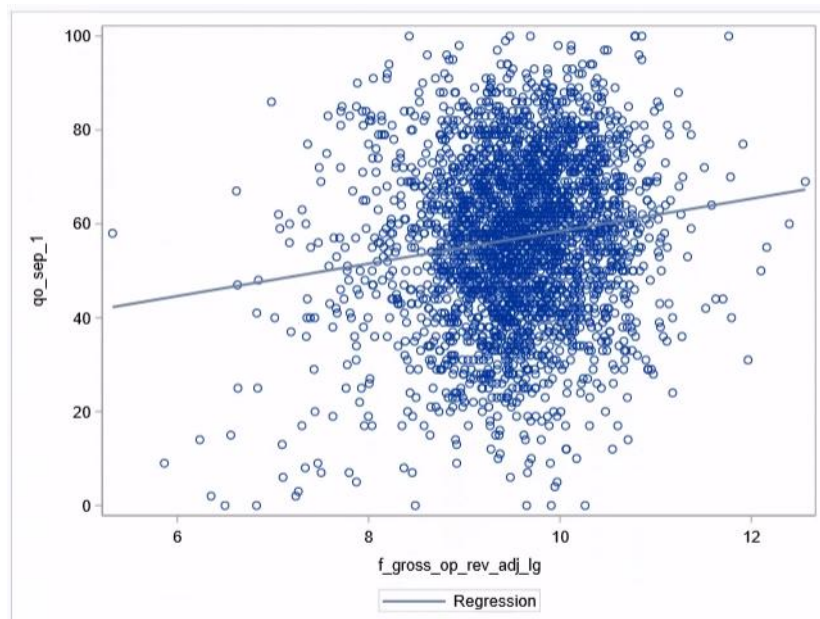
There does not appear to be any correlation between gross outpatients revenue and the death rate of stroke patients.

f_gross_op_rev_adj_lg vs qo_pc_01_lg



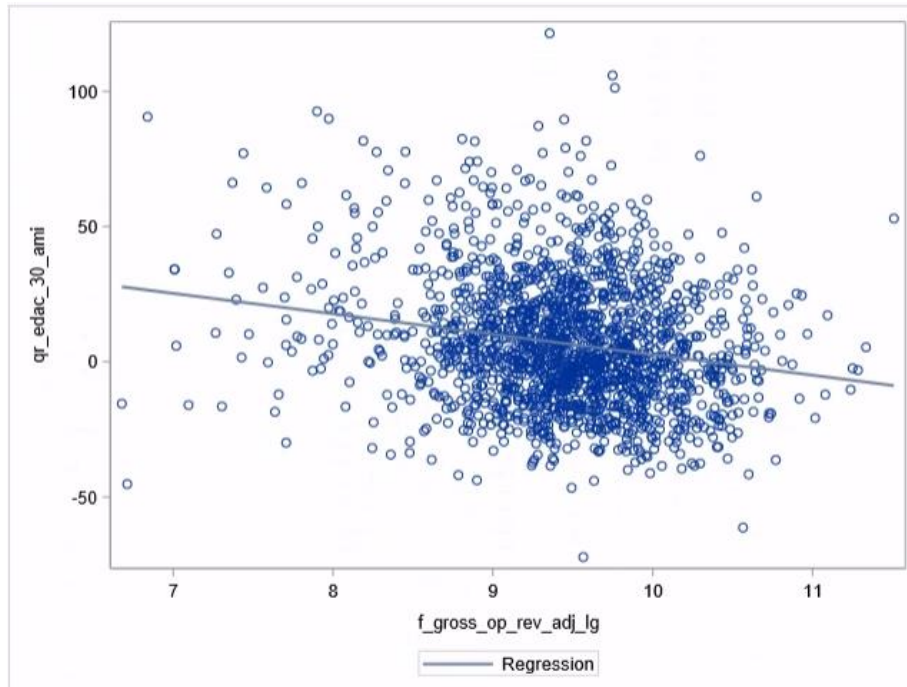
There appears to be a positive relationship between gross outpatient revenue and the percent of mothers whose deliveries were scheduled too early.

f_gross_op_rev_adj_lg vs qo_sep_1



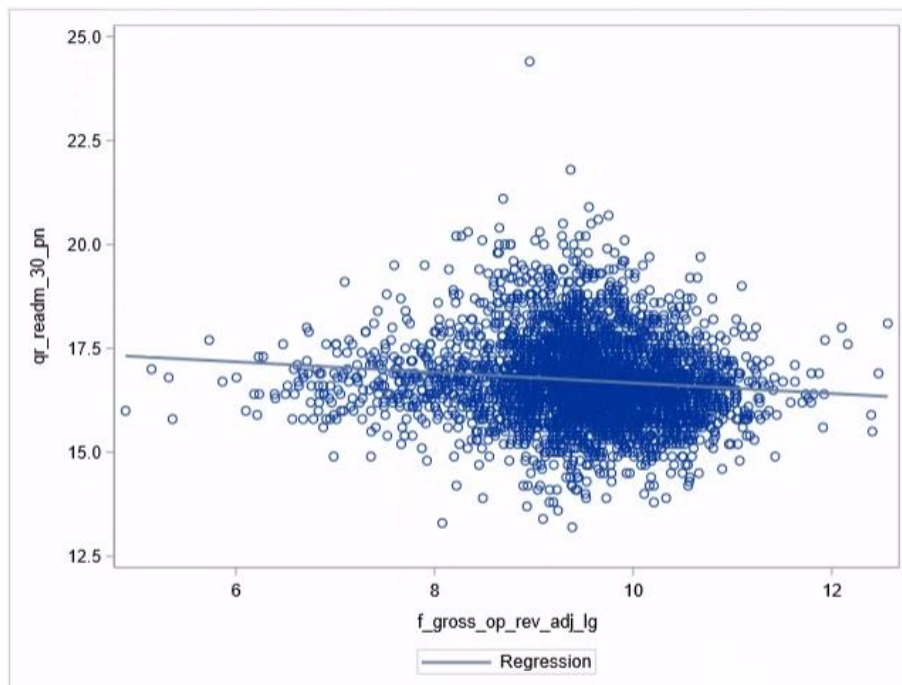
As gross outpatient revenue increases, severe sepsis and septic shock appears to increase.

f_gross_op_rev_adj_lg vs qr_edac_30_ami



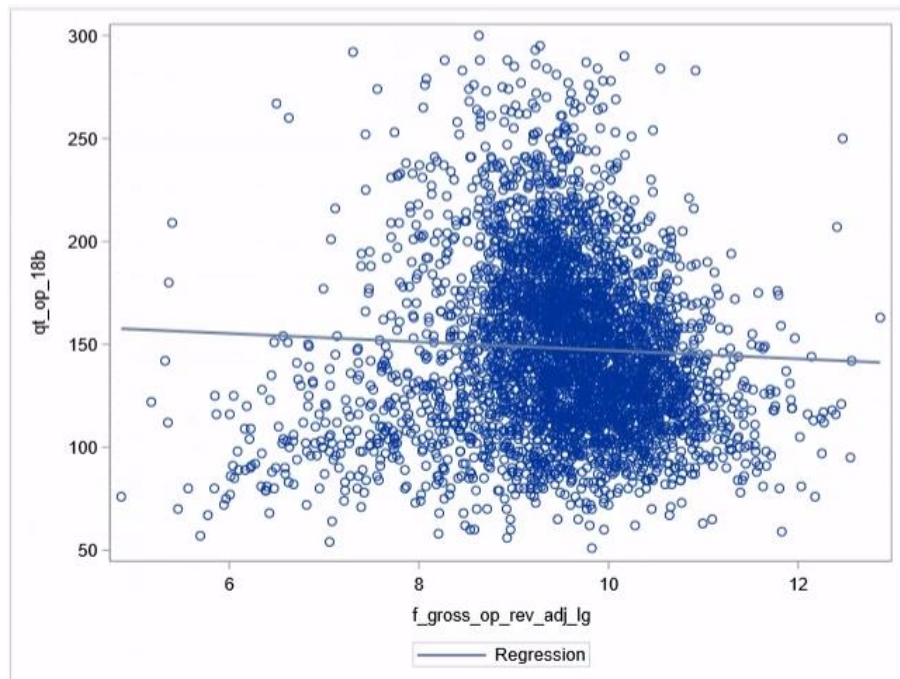
As gross outpatient revenue increases, hospital return days for heart attack patients appear to decrease.

f_gross_op_rev_adj_lg vs qr_readm_30_pn



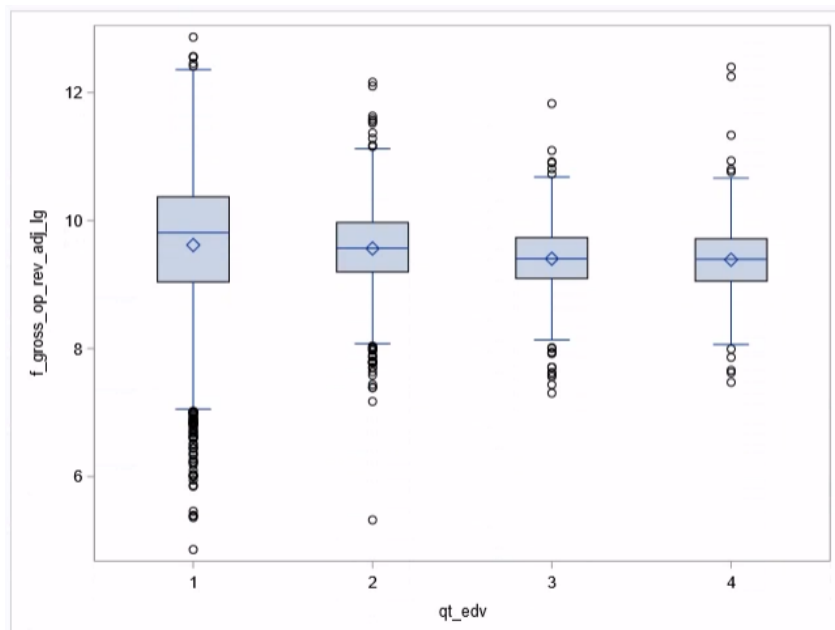
There appears to be a negative relationship between gross outpatient revenue and the readmission rate of pneumonia patients.

f_gross_op_rev_adj_lg vs qt_op_18b



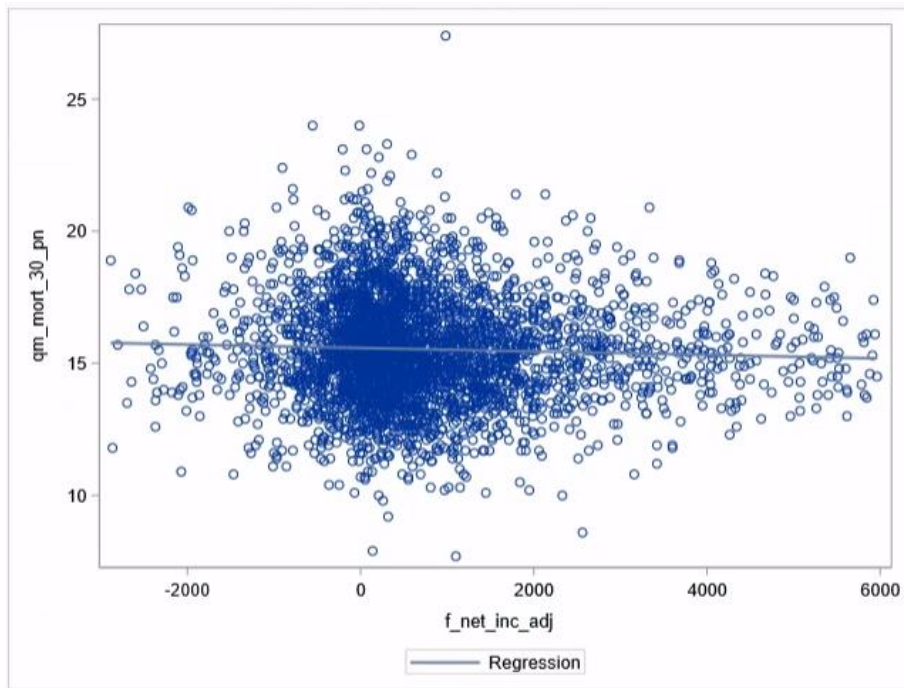
There appears to be a negative relationship between gross outpatient revenue and the median time patients spent in the emergency department before leaving from the visit. There appears to be less variability in the median time when gross outpatient revenue is low.

f_gross_op_rev_adj_lg vs qt_edv



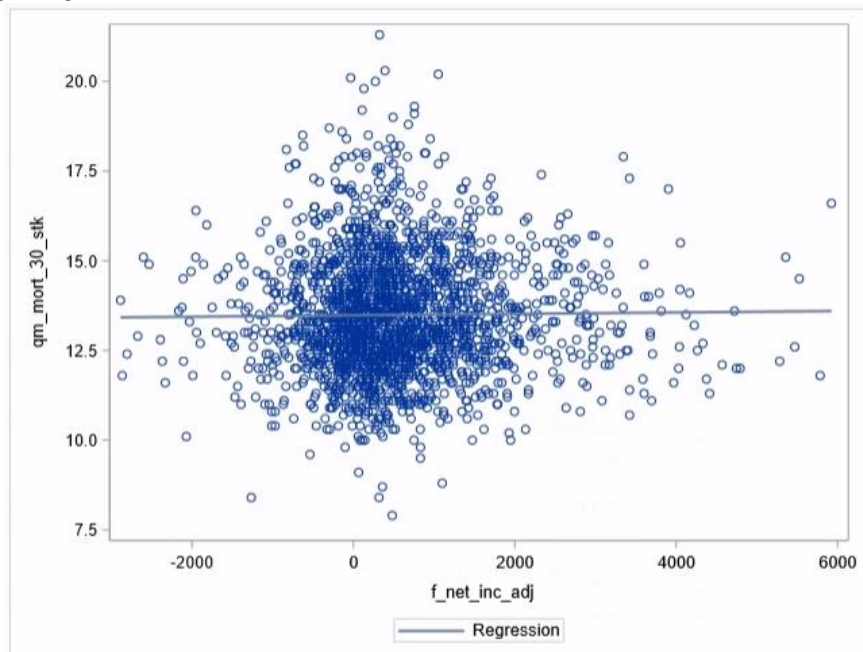
Gross outpatient revenue appears to be the highest when emergency department volume is lowest. That also appears to be where the most variability in outpatient revenue occurs.

f_net_inc_adj vs qm_mort_30_pn



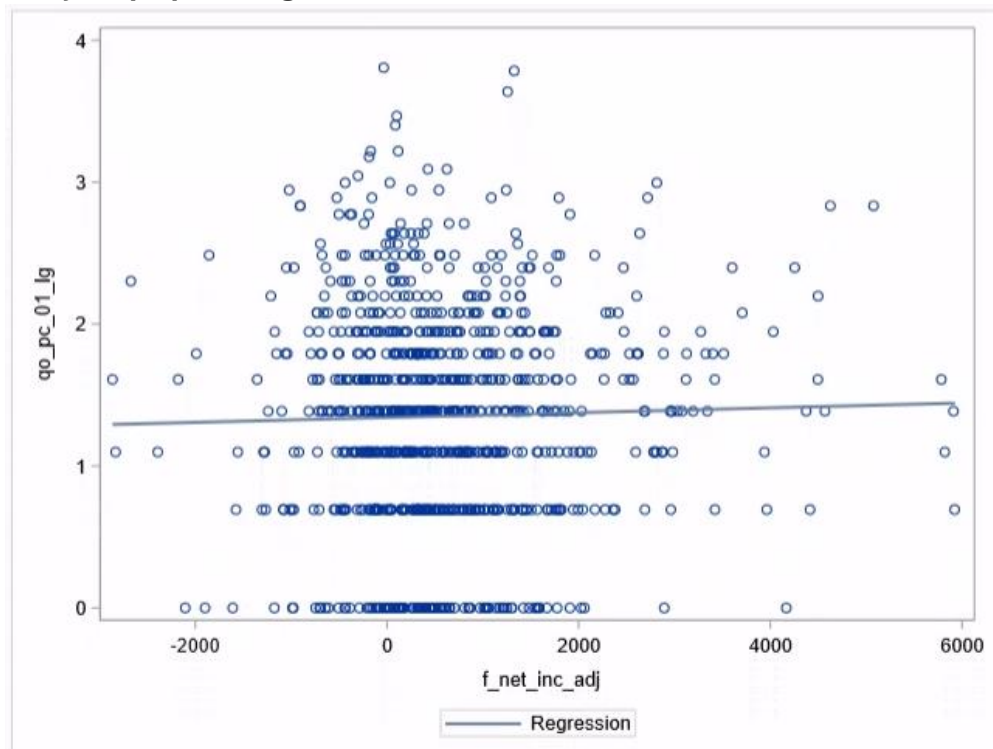
There appears to be a slight negative relationship between net income and the death rate of pneumonia patients. There appears to be less variability in the death rate of pneumonia patients when net income is high.

f_net_inc_adj vs qm_mort_30_stk



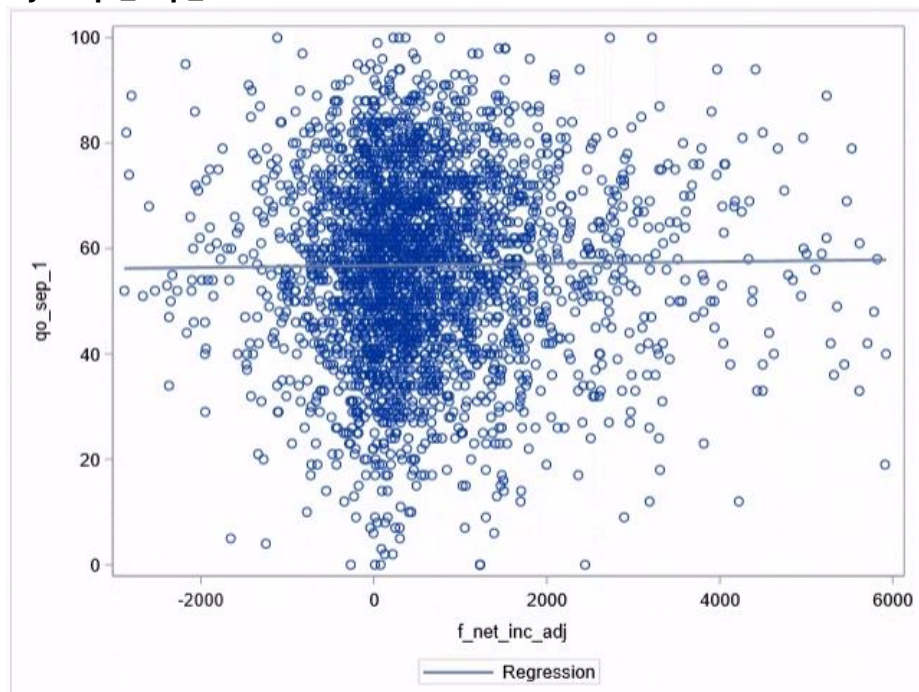
There does not appear to be much correlation between net income and the death rate of stroke patients.

f_net_inc_adj vs qo_pc_01_lg



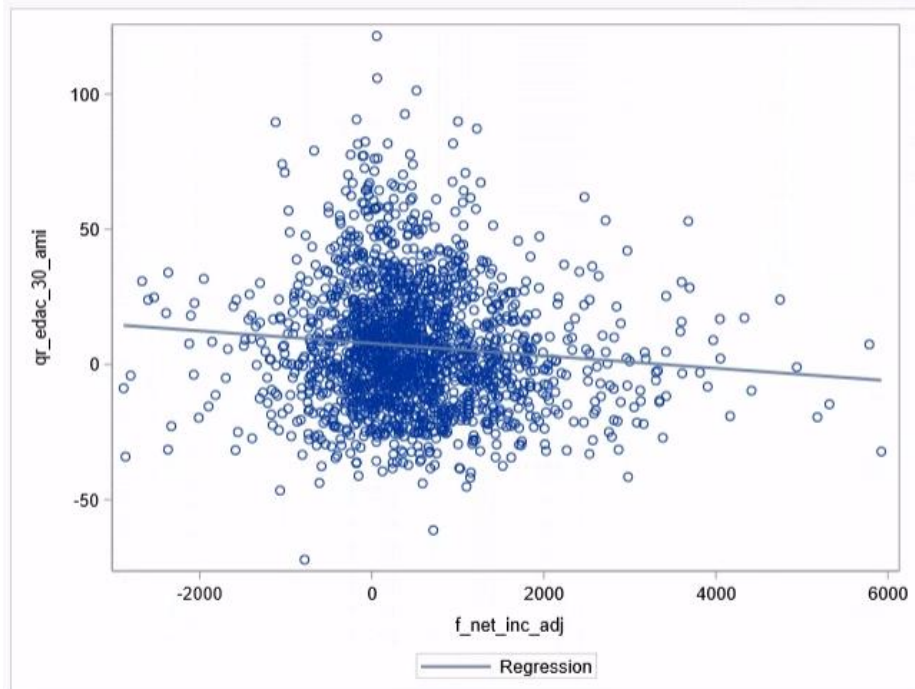
There appears to be a slightly positive relationship between net income and the percentage of mothers whose deliveries were scheduled too early.

f_net_inc_adj vs qo_sep_1



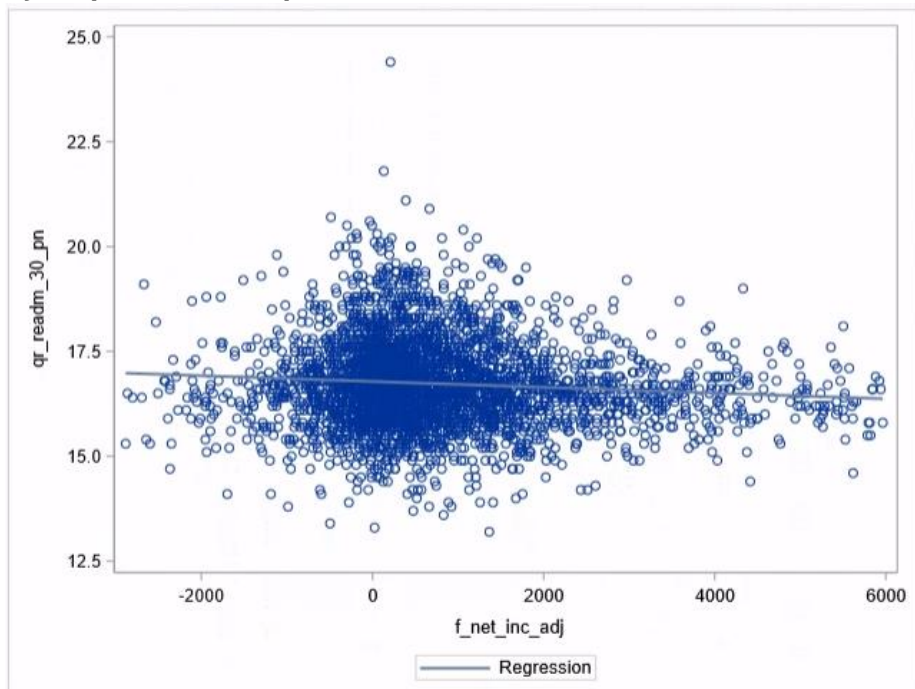
There does not appear to be much correlation between net income and severe sepsis/septic shock.

f_net_inc_adj vs qr_edac_30_ami



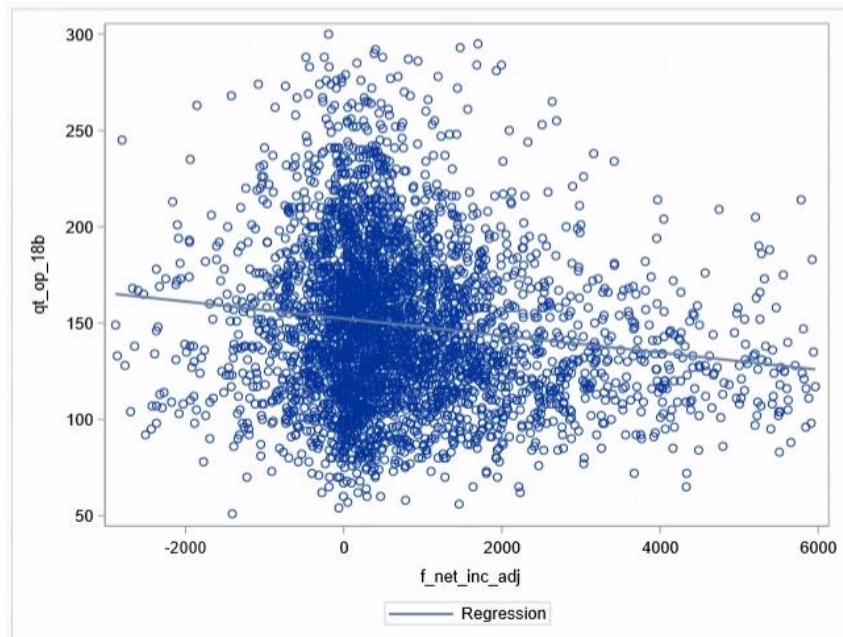
There appears to be a negative relationship between net income and hospital return days for heart attack patients. There appears to be less variability in hospital return days when net income is high.

f_net_inc_adj vs qr_readm_30_pn



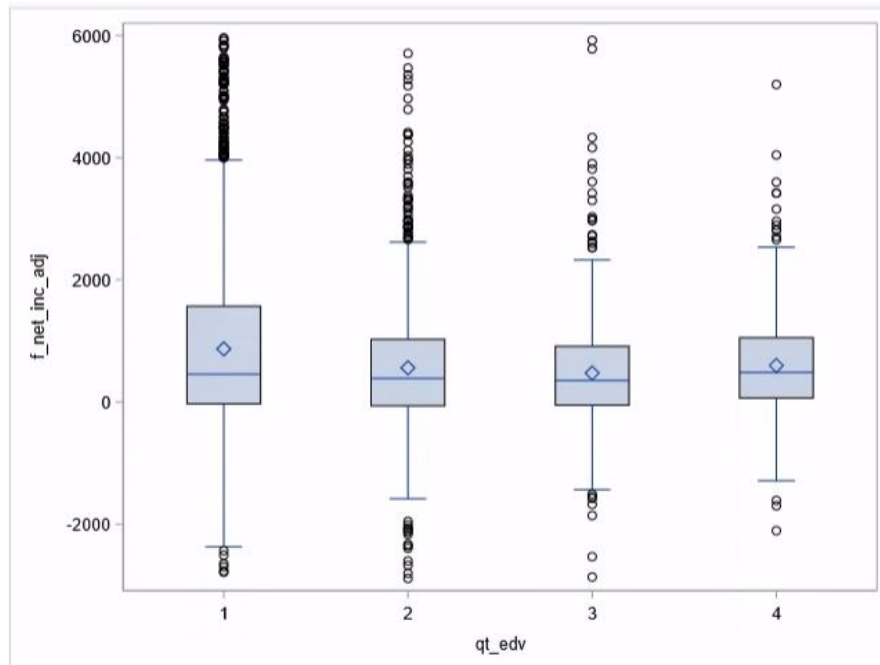
As net income increases, the readmission rate for pneumonia patients appears to decrease.

f_net_inc_adj vs qt_op_18b



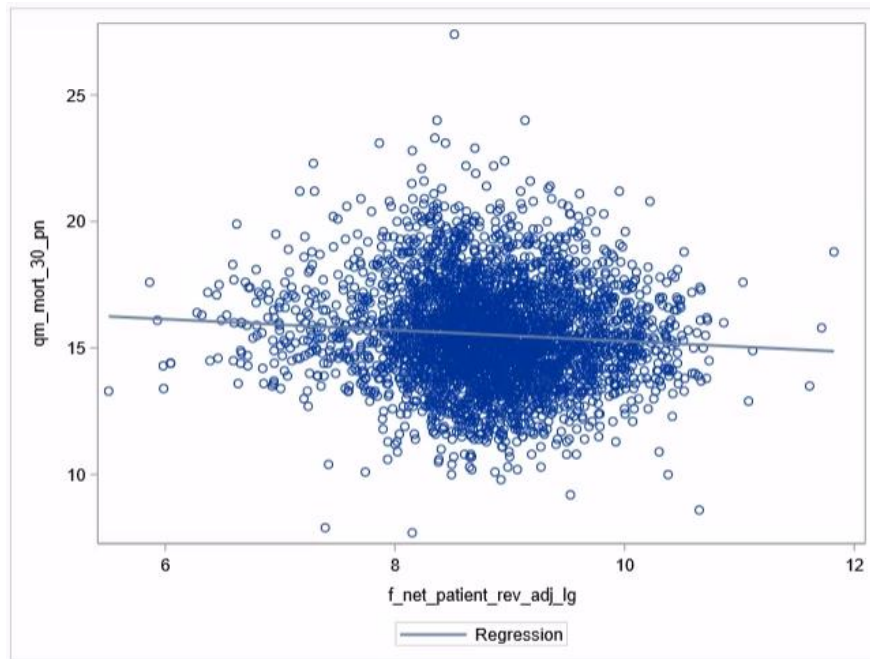
There appears to be a negative relationship between net income and the median time patients spent in the emergency department before leaving from the visit. There is lots of variability in median time, but there is less variability when net income is high.

f_net_inc_adj vs qt_edv



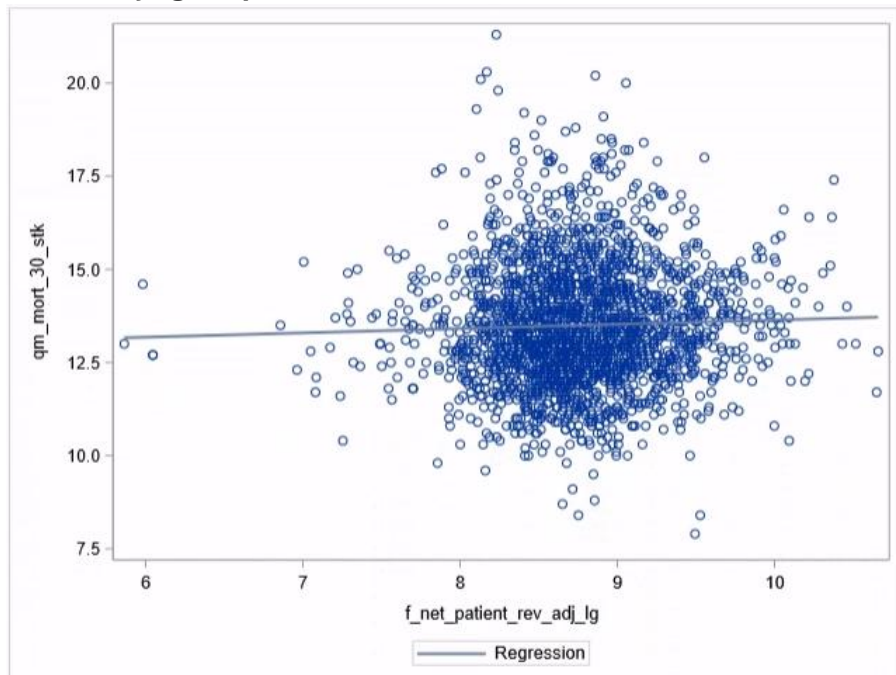
The different emergency department volumes all show pretty similar net income averages. There is more variability in net income when emergency department volume is lower.

f_net_patient_rev_adj_lg vs qm_mort_30_pn



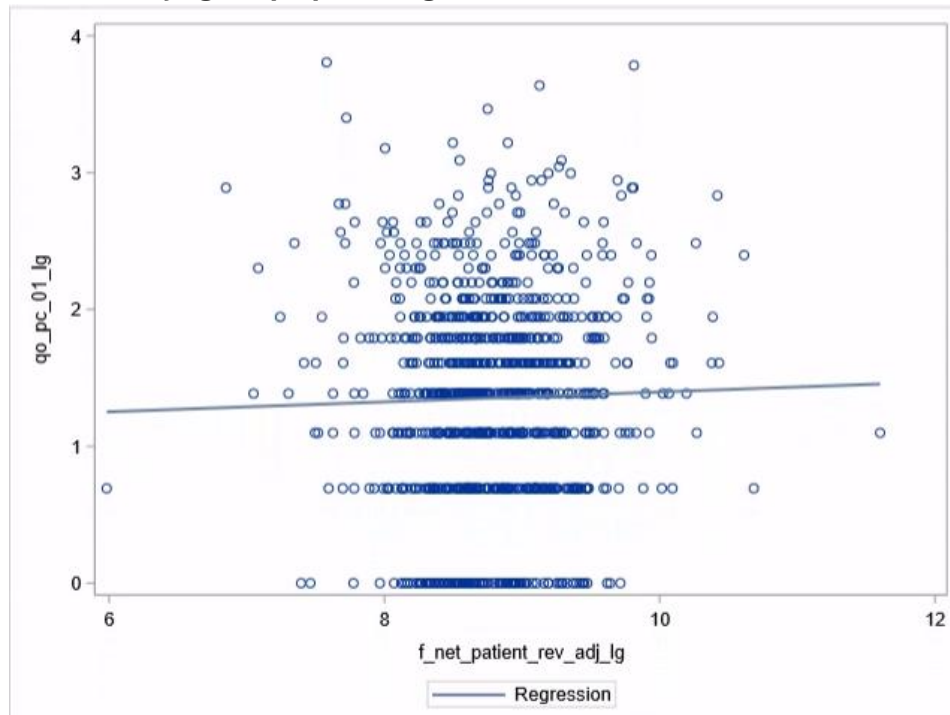
There appears to be a negative relationship between net patient revenue and the death rate for pneumonia patients. There appears to be more variability in the death rate for pneumonia patients when net patient revenue is in the middle range.

f_net_patient_rev_adj_lg vs qm_mort_30_stk



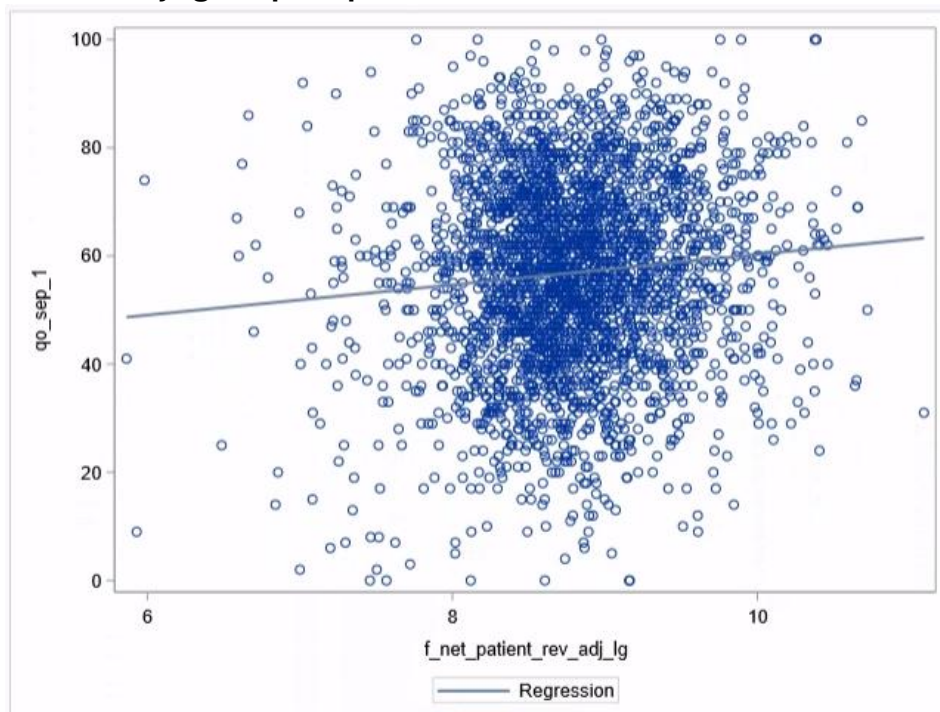
There appears to be a positive relationship between net patient revenue and the death rate of stroke patients.

f_net_patient_rev_adj_lg vs qo_pc_01_lg



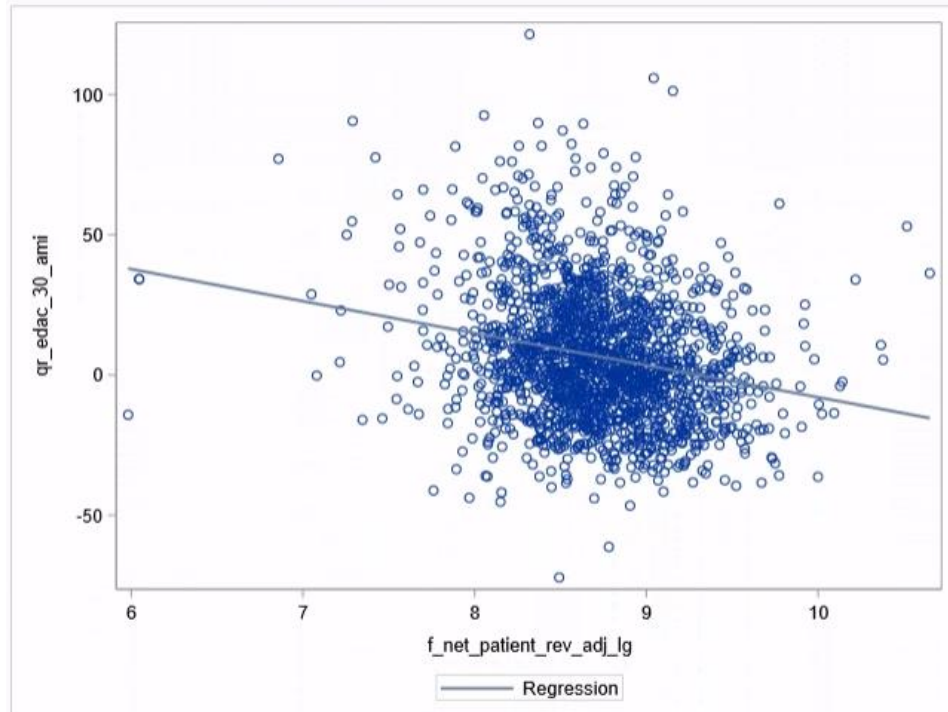
There appears to be a positive relationship between net patients revenue and the percent of mothers whose deliveries were scheduled too early.

f_net_patient_rev_adj_lg vs qo_sep_1



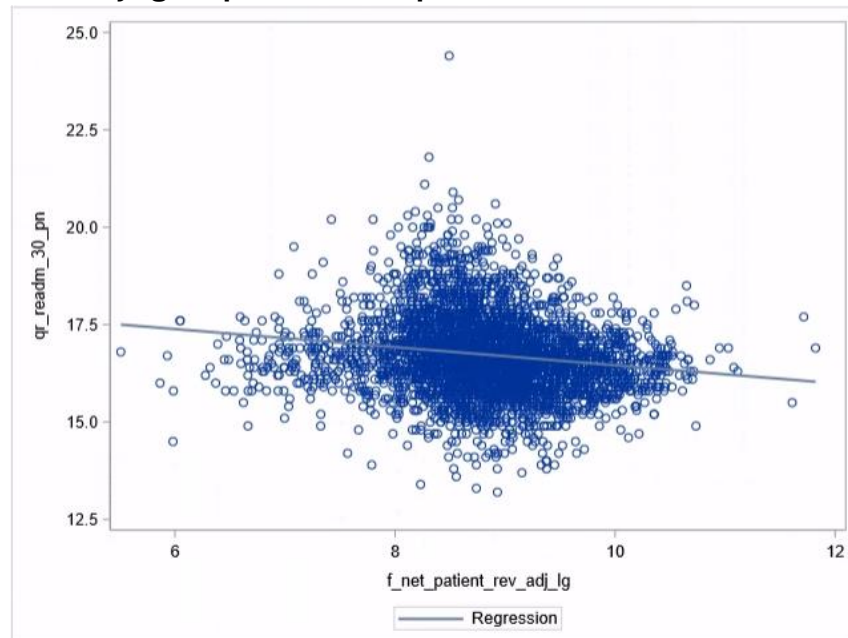
There is appears to be a positive relationship between net patient revenue and severe sepsis/septic shock. There is lots of variability the severe sepsis and septic shock.

f_net_patient_rev_adj_lg vs qr_edac_30_ami



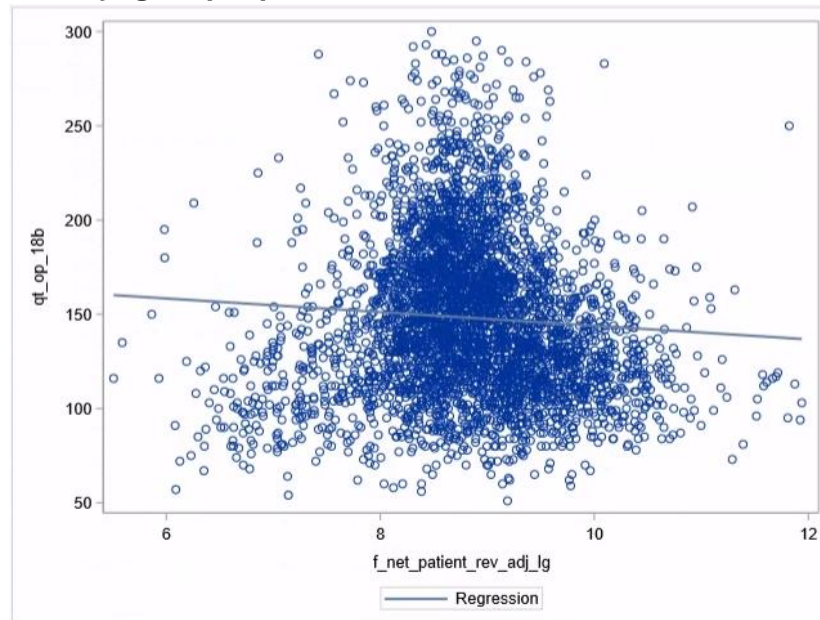
There appears to be a negative relationship between net patient revenue and hospital return days for heart attack patients.

f_net_patient_rev_adj_lg vs qr_readm_30_pn



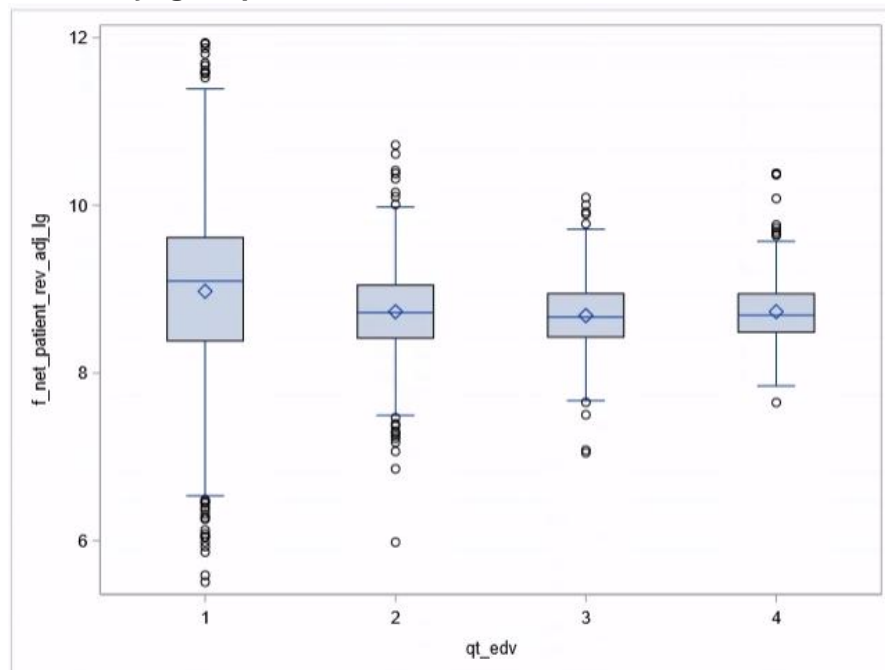
As net patient revenue increases, the readmission rate of pneumonia patients appears to decrease. The readmission rate has the most variability when net patient revenue is in the middle ranges.

f_net_patient_rev_adj_lg vs qt_op_18b



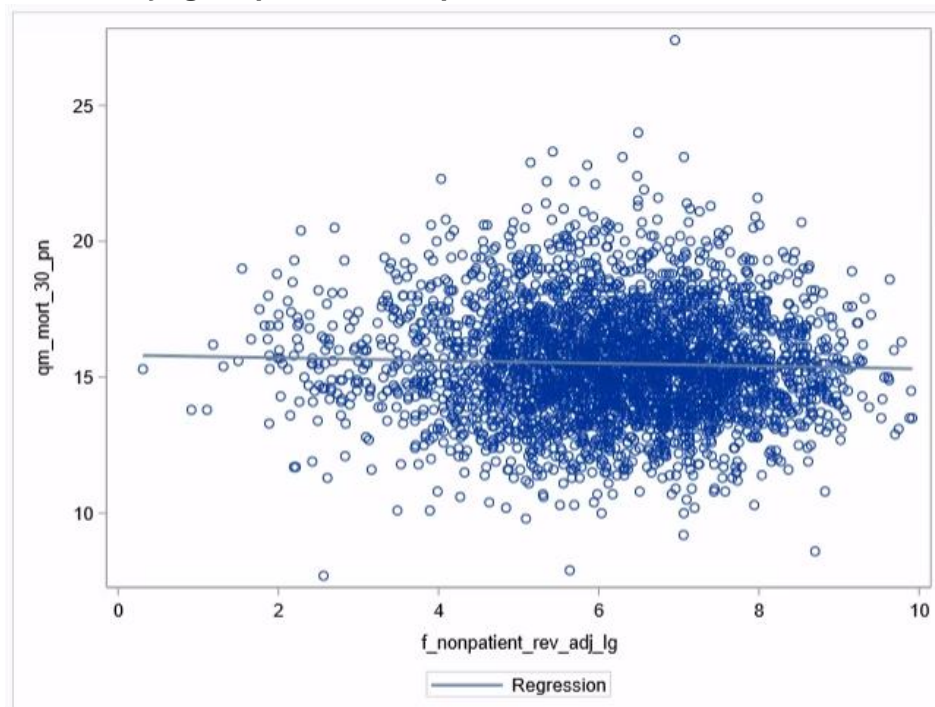
There appears to be a negative relationship between net patient revenue and the median time patients spent in the emergency department before leaving from the visit. There is significant variation in the time patients spent in the emergency department in the middle ranges of net patient revenue.

f_net_patient_rev_adj_lg vs qt_edv



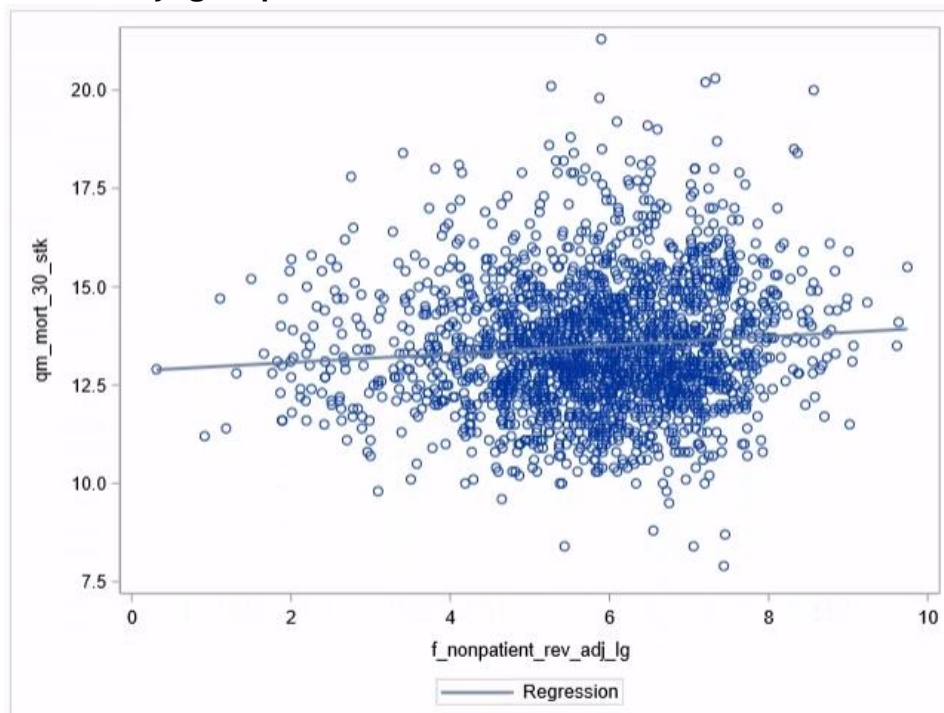
Hospitals with the lowest emergency department volume appear to have the highest average net patient revenue. These hospitals also appear to have the most variability in net patient revenue.

f_nonpatient_rev_adj_lg vs qm_mort_30_pn



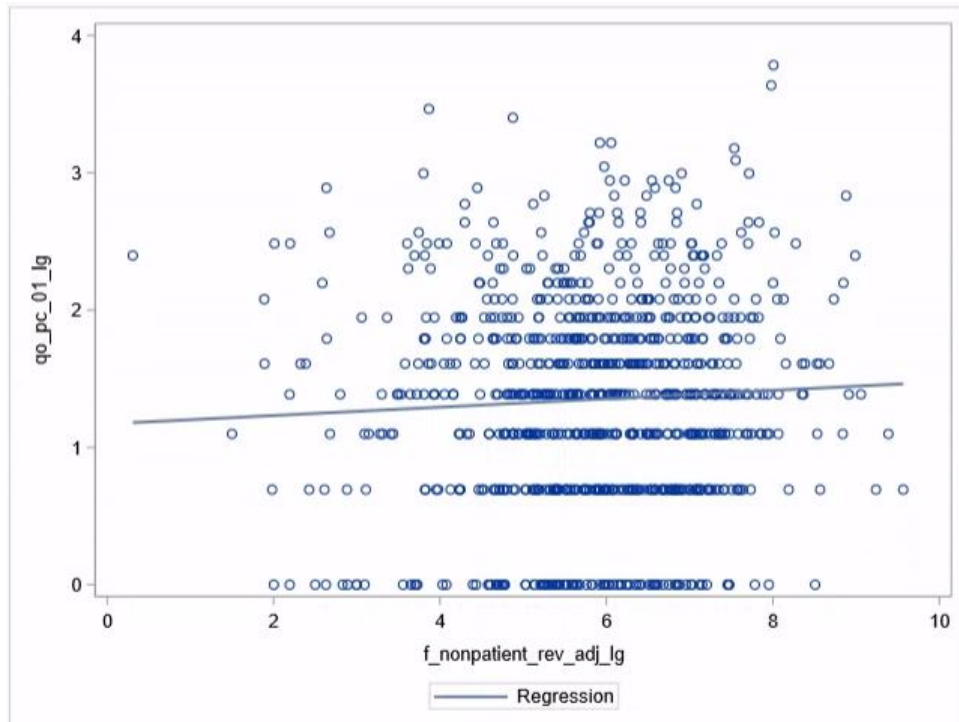
There does not appear to be much correlation between nonpatient revenue and the death rate of pneumonia patients.

f_nonpatient_rev_adj_lg vs qm_mort_30_stk



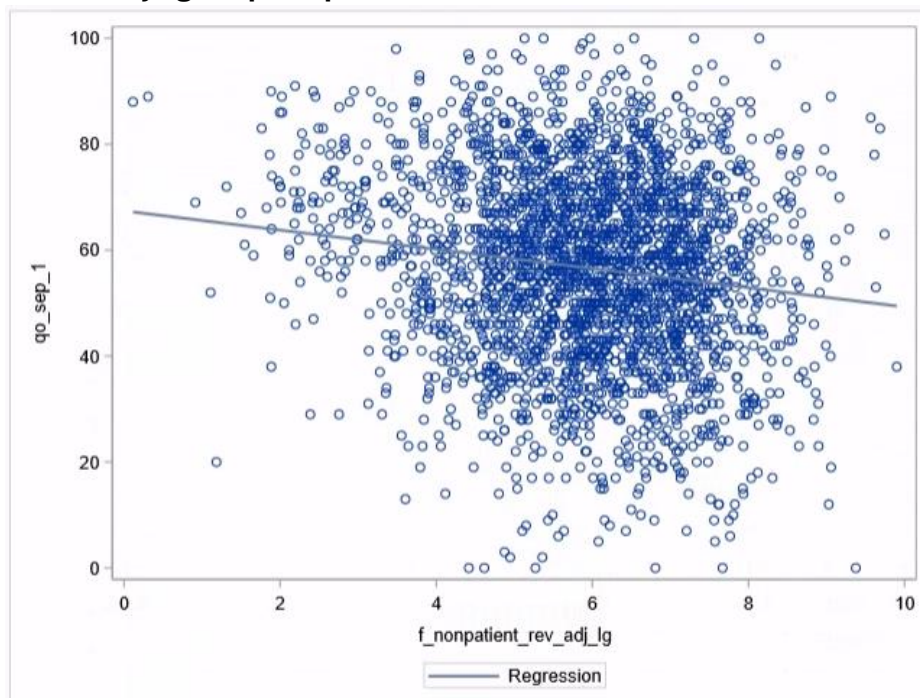
There appears to be a positive relationship between nonpatient revenue and the death rate of stroke patients.

f_nonpatient_rev_adj_lg vs qo_pc_01_lg



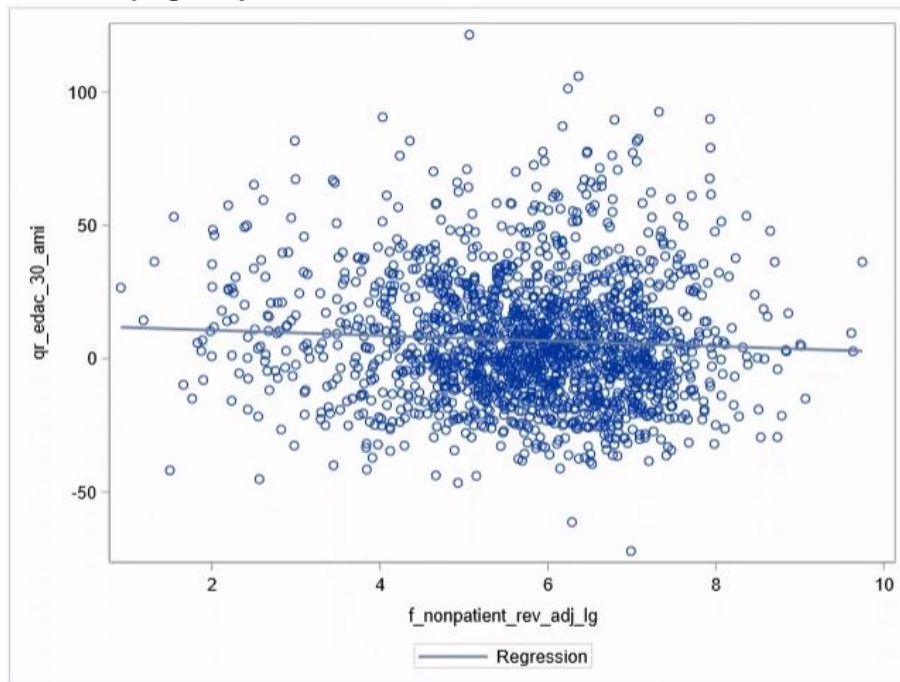
As nonpatient revenue increases, the percent of mothers whose deliveries were scheduled too early appears to increase.

f_nonpatient_rev_adj_lg vs qo_sep_1



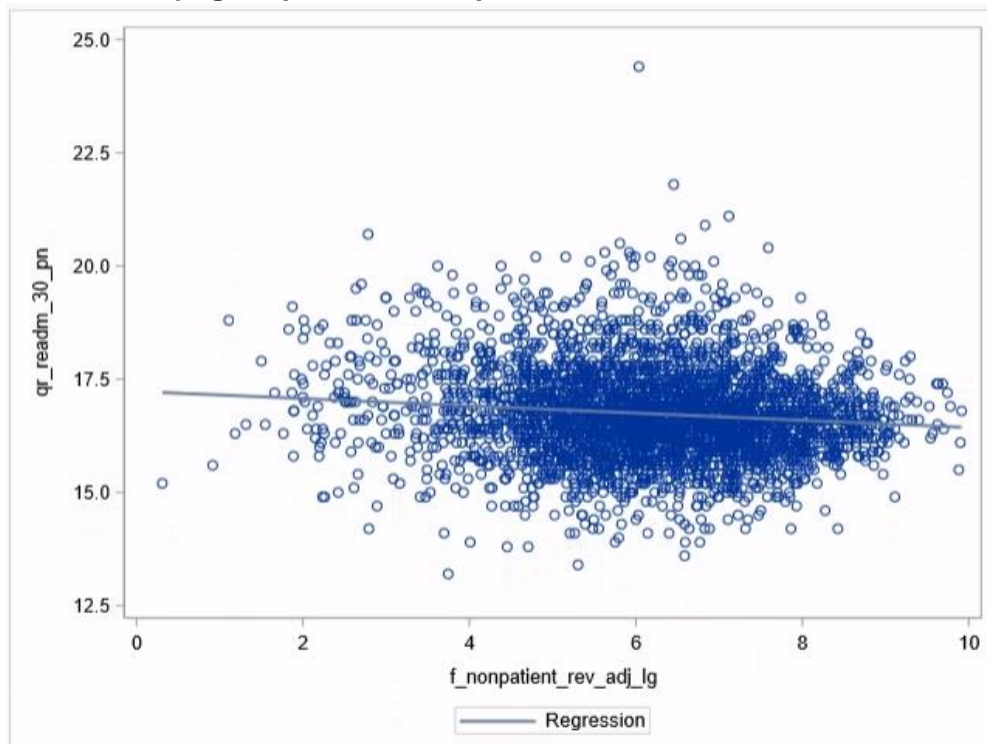
There appears to be a negative relationship between nonpatient revenue and severe sepsis/septic shock. There is lots of variability in severe sepsis and septic shock.

f_nonpatient_rev_adj_lg vs qr_edac_30_ami



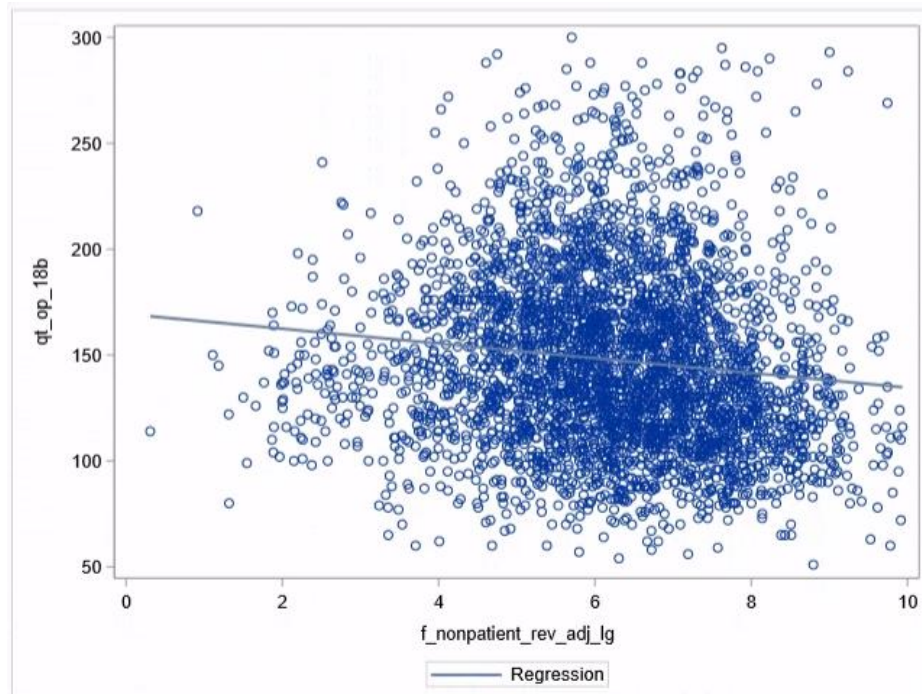
There appears to be a slight negative relationship between nonpatient revenue and hospital return days for heart attack patients.

f_nonpatient_rev_adj_lg vs qr_readm_30_pn



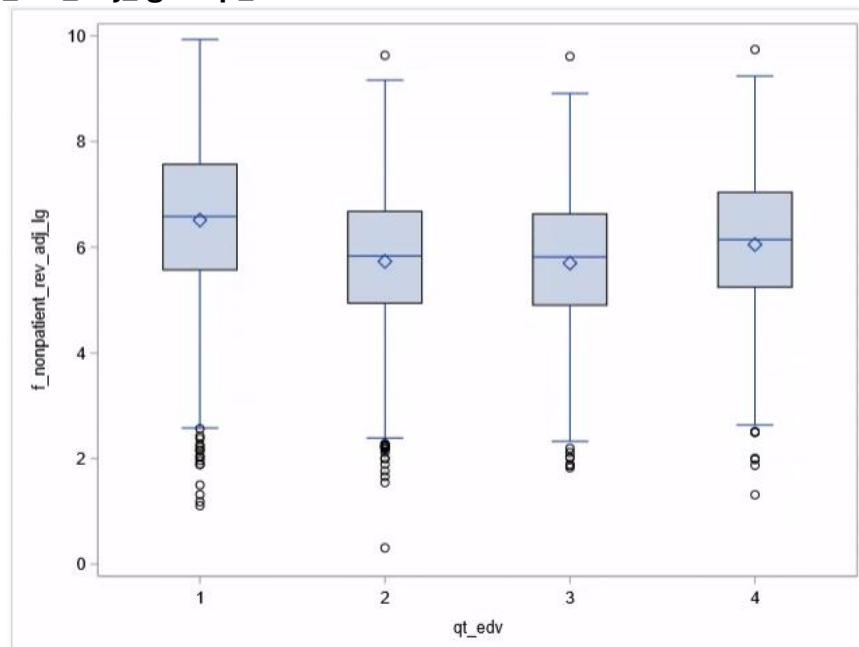
There appears to be a negative relationship between nonpatient revenue and the readmission rate of pneumonia patients.

f_nonpatient_rev_adj_lg vs qt_op_18b



There appears to be a negative relationship between nonpatient revenue and the median time patients spent in the emergency department before leaving from the visit. There appears to be less variability in the time patients spent when nonpatient revenue is lower.

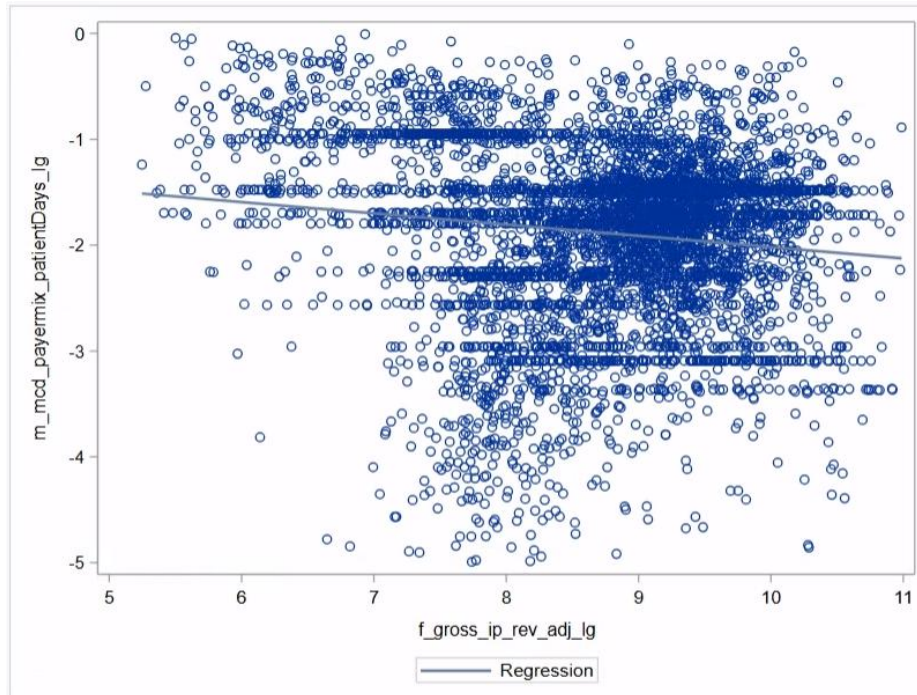
f_nonpatient_rev_adj_lg vs qt_edv



Hospitals with the lowest emergency department volume appear to have the highest average nonpatient revenue.

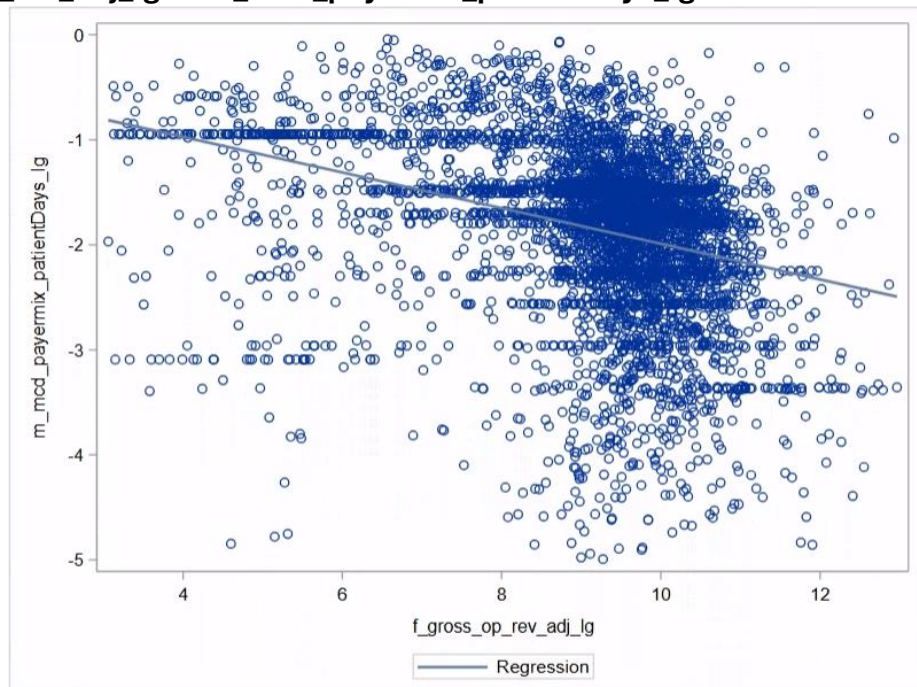
Finance vs Medicare and Medicaid:

f_gross_ip_rev_adj_lg vs m_mcd_payermix_patientDays_lg



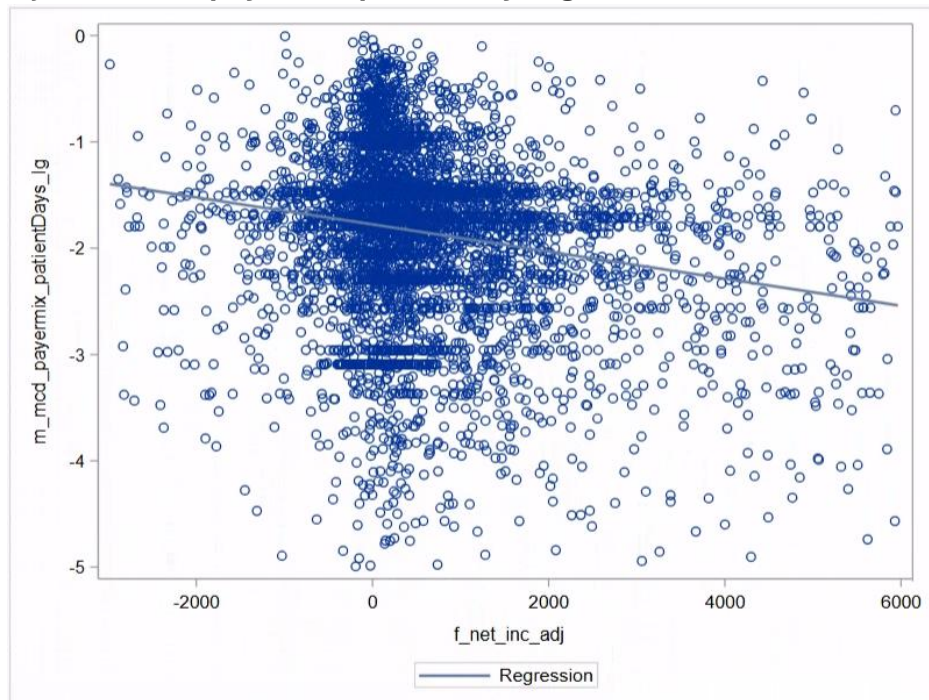
There appears to be a negative relationship between gross inpatient revenue and Medicaid payer mix.

f_gross_op_rev_adj_lg vs m_mcd_payermix_patientDays_lg



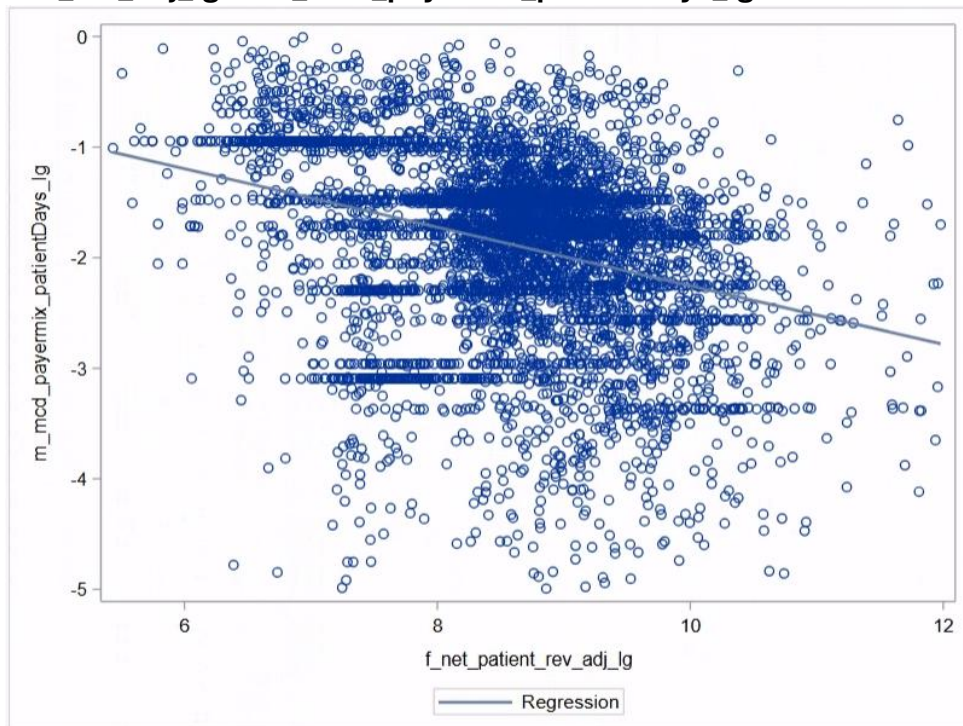
As gross outpatient revenue increases, Medicaid payer mix appears to decrease. There is lots of variability in Medicaid payer mix throughout the graph.

f_net_inc_adj vs m_mcd_payermix_patientDays_lg



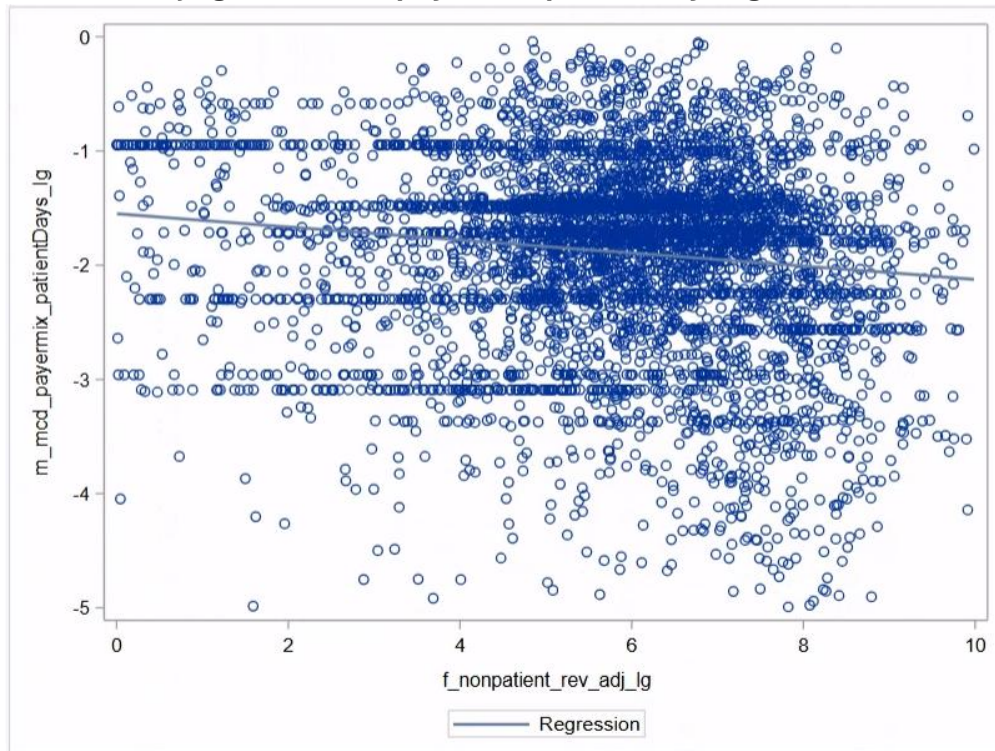
There appears to be a negative relationship between net income and Medicaid payer mix.

f_net_patient_rev_adj_lg vs m_mcd_payermix_patientDays_lg



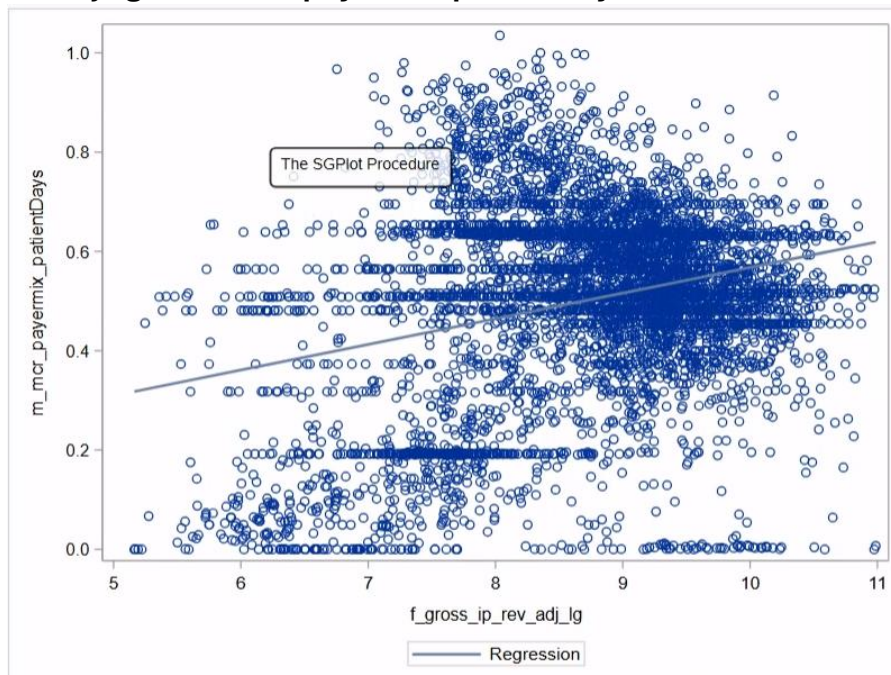
There appears to be a negative relationship between net patient revenue and Medicaid payer mix.

f_nonpatient_rev_adj_lg vs m_mcd_payermix_patientDays_lg



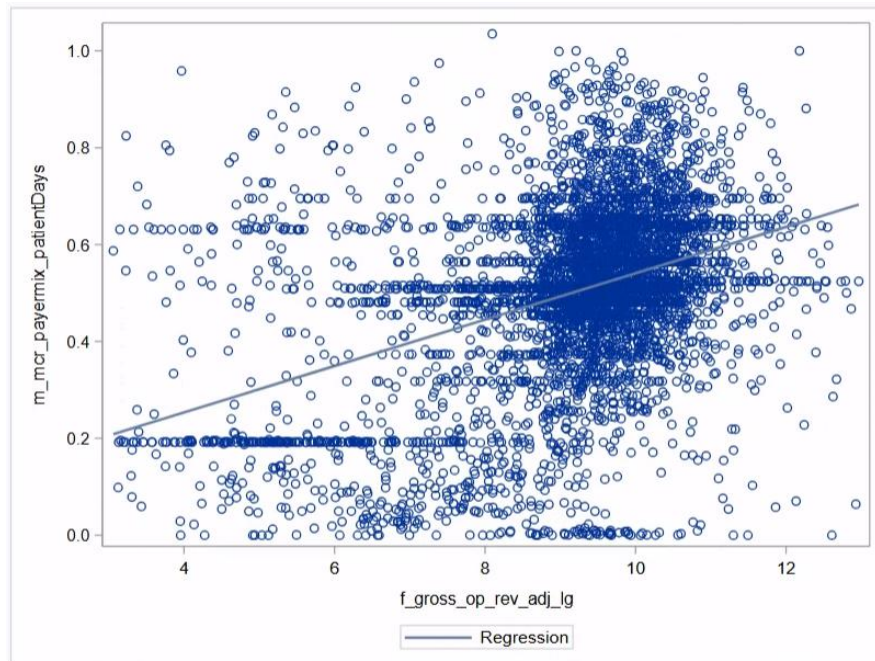
There appears to be a slight negative relationship between nonpatient revenue and Medicaid payer mix.

f_gross_ip_rev_adj_lg vs m_mcr_payermix_patientDays



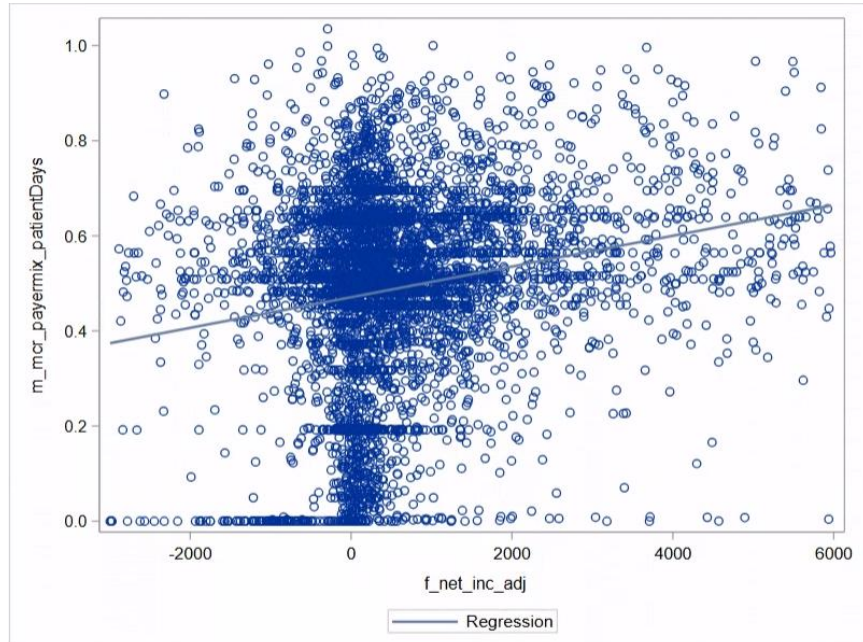
There appears to be a positive relationship between gross inpatient revenue and Medicare payer mix

f_gross_op_rev_adj_lg vs m_mcr_payermix_patientDays



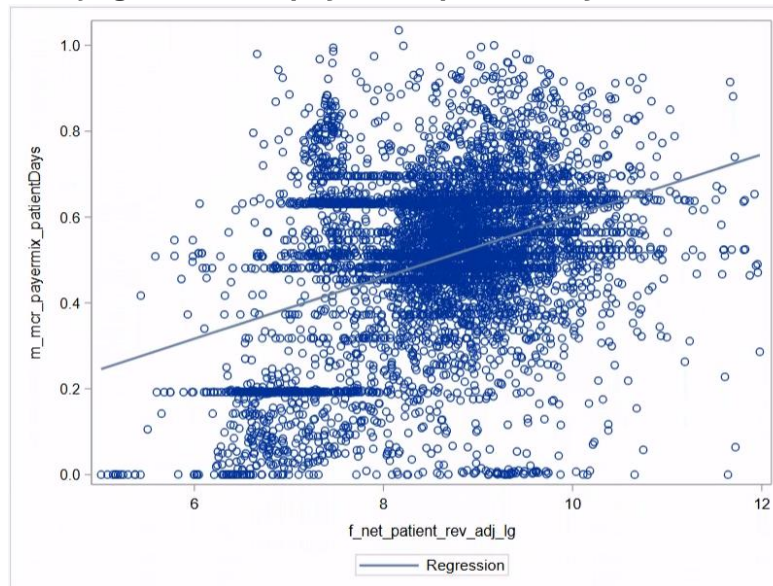
There appears to be a positive relationship between gross outpatient revenue and Medicare payer mix.

f_net_inc_adj vs m_mcr_payermix_patientDays



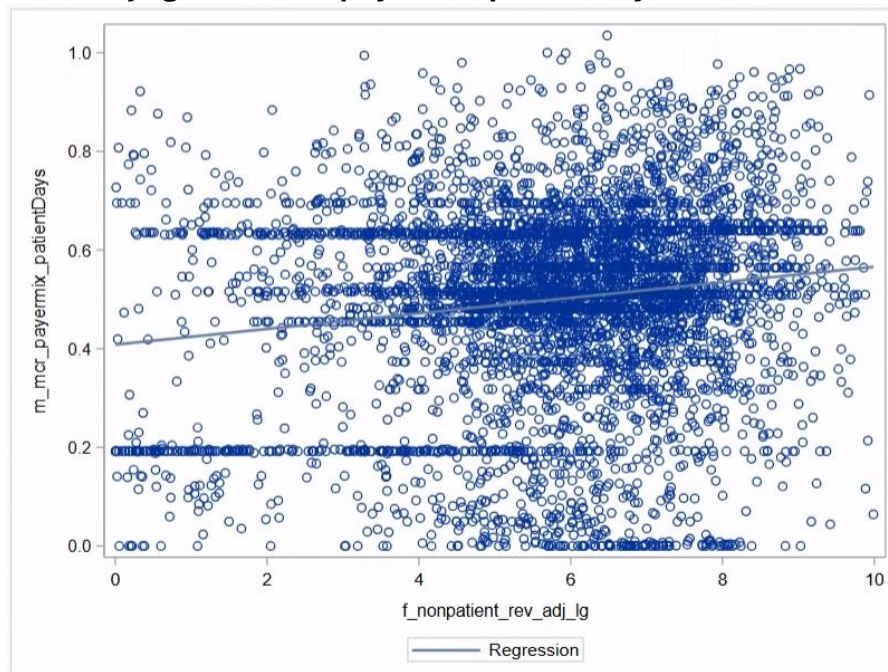
There appears to be a positive relationship between net income and Medicare payer mix. There appears to be lots of variation in Medicare payer mix when net income is close to 0.

f_net_patient_rev_adj_lg vs m_mcr_payermix_patientDays



There appears to be a positive relationship between net patient revenue and Medicare payer mix.

f_nonpatient_rev_adj_lg vs m_mcr_payermix_patientDays



There appears to be a slight positive relationship between nonpatient revenue and Medicare payer mix.

Hypothesis:

As the readmission rate for pneumonia patients increases, net income decreases.

As the median time patients spent in the emergency department before leaving from the visit increases, gross inpatient revenue increases.

Hospitals with the lower emergency department volume have lower gross inpatient revenue.

As Medicare payer mix increases, net income increases.

Testing Section

As the readmission rate for pneumonia patients increases, net income decreases.

Linear Regression Results	
The REG Procedure	
Model: Linear_Regression_Model	
Dependent Variable: f_net_inc_adj	
Number of Observations Read	4575
Number of Observations Used	2873
Number of Observations with Missing Values	1702

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	14	591011015	42215072	25.41	<.0001
Error	2858	4747941477	1661281		
Corrected Total	2872	5338952492			

Root MSE	1288.90695	R-Square	0.1107
Dependent Mean	785.45968	Adj R-Sq	0.1063
Coeff Var	164.09588		

This regression includes hospitals with less than 300 beds that have a net income between -3000 and 6000 per patient day. The distribution of residuals graph looks good enough to proceed. The model does significantly explain 11.07% of net income per patient day.

Parameter Estimates						
Variable	Label	DF	Parameter Estimate	Standard Error	t Value	Pr > t
Intercept	Intercept	1	1937.21704	448.46653	4.32	<.0001
qr_readm_30_pn		1	-61.11817	25.43375	-2.40	0.0163
d_beds	BEDS, TOTAL (BY TYPE)	1	-3.09305	0.38304	-8.08	<.0001
d_CBSA_micro		1	17.50595	66.80837	0.26	0.7933
d_CBSA_rural		1	-46.47171	71.46263	-0.65	0.5156
d_FP		1	205.10328	72.74348	2.82	0.0048
d_GOV		1	176.88683	64.85180	2.73	0.0064
d_COTH		1	416.91798	326.18024	1.28	0.2013
d_r_northeast		1	-144.63050	95.58184	-1.51	0.1303
d_r_southcentral		1	-192.34845	81.13645	-2.37	0.0178
d_r_northcentral		1	289.89147	76.32177	3.80	0.0001
d_r_mountain		1	422.58926	105.51674	4.00	<.0001
d_r_pacific		1	-103.58038	96.30109	-1.08	0.2822
m_mcr_payermix_patientDays		1	498.34734	201.20202	2.48	0.0133
m_mcd_payermix_patientDays		1	-1061.88633	262.04257	-4.05	<.0001

For hospitals in the regression, an increase of one in the readmission rate for pneumonia patients is correlated with a decrease of 61.12 dollars in net income per patient day. An increase of one bed is correlated with a decrease of 3.09 dollars in net income per patient day. Not for profit hospitals see 205.10 dollars less in net income per patient day than for profit hospitals and 176.89 dollars less in net income per patient day than government hospitals. Hospitals in the southeast see 192.35 dollars more than hospitals in the south central region, 289.89 dollars less than hospitals in the north central region, and 422.59 dollars less than hospitals in the mountain region in net income per patient day. An increase of 10% in Medicare payer mix per patient day is correlated with an increase of 49.83 dollars in net income per patient day. An increase of 10% in Medicaid payer mix per patient day is correlated with a decrease of 106.19 dollars in net income per patient day.

As the median time patients spent in the emergency department before leaving from the visit increases, gross inpatient revenue increases.

Linear Regression Results					
The REG Procedure					
Model: Linear_Regression_Model					
Dependent Variable: f_gross_ip_rev_adj_lg					
Number of Observations Read					4710
Number of Observations Used					3154
Number of Observations with Missing Values					1556

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	14	1448.92002	103.49429	196.86	<.0001
Error	3139	1650.28471	0.52574		
Corrected Total	3153	3099.20473			

Root MSE	0.72508	R-Square	0.4675
Dependent Mean	8.73205	Adj R-Sq	0.4651
Coeff Var	8.30362		

This regression includes hospitals with less than 300 beds that have a logged gross inpatient revenue per patient day between 5 and 11. The distribution of residuals graph looks good enough to proceed. The model does significantly explain 46.75% of logged gross inpatient revenue per patient day.

Parameter Estimates						
Variable	Label	DF	Parameter Estimate	Standard Error	t Value	Pr > t
Intercept	Intercept	1	7.93218	0.10325	76.82	<.0001
qt_op_18b		1	0.00496	0.00043098	11.51	<.0001
d_beds	BEDS, TOTAL (BY TYPE)	1	0.00152	0.00021832	6.98	<.0001
d_CBSA_micro		1	-0.12052	0.03572	-3.37	0.0007
d_CBSA_rural		1	-0.71937	0.03752	-19.17	<.0001
d_FP		1	0.50514	0.03922	12.88	<.0001
d_GOV		1	-0.19762	0.03450	-5.73	<.0001
d_COTH		1	0.11256	0.17838	0.63	0.5281
d_r_northeast		1	-0.15368	0.05281	-2.91	0.0036
d_r_southcentral		1	0.07425	0.04463	1.66	0.0963
d_r_northcentral		1	-0.11037	0.04216	-2.62	0.0089
d_r_mountain		1	0.25234	0.05539	4.56	<.0001
d_r_pacific		1	0.42938	0.05315	8.08	<.0001
m_mcr_payermix_patientDays		1	0.78838	0.10107	7.80	<.0001
m_mcd_payermix_patientDays		1	-1.69520	0.13365	-12.68	<.0001

For hospitals in the regression, an increase of ten minutes in the median time patients spent in the emergency room before leaving from the visit is correlated with an increase of 0.05 in logged gross inpatient revenue per patient day. Ten additional beds are correlated with an increase of 0.02 in logged gross inpatient revenue per patient day. It seems as though most of the explanatory variables have a statistically significant relationship with the dependent variable. I believe that logging gross inpatient revenue per patient day could have caused this to occur.

Hospitals with the lower emergency department volume have lower gross inpatient revenue.

Linear Regression Results

The REG Procedure
Model: Linear_Regression_Model
Dependent Variable: f_gross_ip_rev_adj_lg

Number of Observations Read	4715
Number of Observations Used	2793
Number of Observations with Missing Values	1922

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	14	1116.43484	79.74535	165.75	<.0001
Error	2778	1336.54095	0.48112		
Corrected Total	2792	2452.97579			

Root MSE	0.69363	R-Square	0.4551
Dependent Mean	8.82764	Adj R-Sq	0.4524
Coeff Var	7.85743		

This regression includes hospitals with less than 300 beds that have a logged gross inpatient revenue per patient day between 5 and 11. The distribution of residuals graph looks good enough to proceed. The model does significantly explain 45.51% of logged gross inpatient revenue per patient day.

Parameter Estimates						
Variable	Label	DF	Parameter Estimate	Standard Error	t Value	Pr > t
Intercept	Intercept	1	8.28455	0.09739	85.07	<.0001
qt_edv		1	0.32613	0.02539	12.84	<.0001
d_beds	BEDS, TOTAL (BY TYPE)	1	0.00013428	0.00025398	0.53	0.5971
d_CBSA_micro		1	-0.10926	0.03574	-3.06	0.0023
d_CBSA_rural		1	-0.69243	0.03954	-17.51	<.0001
d_FP		1	0.51323	0.03849	13.34	<.0001
d_GOV		1	-0.20871	0.03665	-5.70	<.0001
d_COTH		1	0.25041	0.17086	1.47	0.1429
d_r_northeast		1	-0.06213	0.05296	-1.17	0.2408
d_r_southcentral		1	0.07763	0.04452	1.74	0.0813
d_r_northcentral		1	-0.06056	0.04276	-1.42	0.1568
d_r_mountain		1	0.29382	0.05783	5.08	<.0001
d_r_pacific		1	0.47961	0.05337	8.99	<.0001
m_mcr_payermix_patientDays		1	0.67114	0.11024	6.09	<.0001
m_mcd_payermix_patientDays		1	-1.53275	0.14300	-10.72	<.0001

For hospitals included in the regression, an increase of one level in emergency department volume is correlated with an increase of 0.32 in logged gross inpatient revenue per patient day. Several of the secondary variables have very small but statistically significant impacts on logged gross inpatient revenue per patient day.

As Medicare payer mix increases, net income increases.

Linear Regression Results

The REG Procedure
Model: Linear_Regression_Model
Dependent Variable: f_net_inc_adj

Number of Observations Read	4575
Number of Observations Used	4575

Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	13	767022601	59001739	37.72	<.0001
Error	4561	7135004147	1564351		
Corrected Total	4574	7902026748			

Root MSE	1250.74011	R-Square	0.0971
Dependent Mean	629.33083	Adj R-Sq	0.0945
Coeff Var	198.74128		

This regression includes hospitals with less than 300 beds that have a net income between -3000 and 6000 per patient day. The distribution of residuals graph looks good enough to proceed. The model does significantly explain 9.71% of net income per patient day.

Parameter Estimates						
Variable	Label	DF	Parameter Estimate	Standard Error	t Value	Pr > t
Intercept	Intercept	1	163.65878	103.84108	1.58	0.1151
m_mcr_payermix_patientDays		1	1057.88142	113.79900	9.30	<.0001
d_beds	BEDS, TOTAL (BY TYPE)	1	-1.96719	0.28833	-6.82	<.0001
d_CBSA_micro		1	164.00172	55.15577	2.97	0.0030
d_CBSA_rural		1	182.72208	55.72509	3.28	0.0010
d_FP		1	-96.29425	47.17798	-2.04	0.0413
d_GOV		1	22.46970	52.84110	0.43	0.6707
d_COTH		1	636.94848	253.25838	2.52	0.0119
d_r_northeast		1	-160.72897	73.46984	-2.19	0.0287
d_r_southcentral		1	-42.23265	60.36120	-0.70	0.4842
d_r_northcentral		1	311.27769	59.45464	5.24	<.0001
d_r_mountain		1	362.03246	78.38915	4.62	<.0001
d_r_pacific		1	31.60871	75.27924	0.42	0.6746
m_mcd_payermix_patientDays		1	-135.36654	154.19057	-0.88	0.3800

For hospitals included in this regression, an increase of 10% in Medicare payer mix per patient day is correlated with a 105.79 dollar increase in net income per patient day. An increase of one bed is correlated with a decrease of 1.97 dollars in net income per patient day. Metro hospitals see 164 dollars less than micro hospitals and 182.72 dollars less than rural hospitals in net income per patient day. For profit hospitals see 96.29 dollars less than not for profit hospitals in net income per patient day. Hospitals that are members of the Council of Teaching Hospitals see 636.95 dollars more in net income per patient day than hospitals that are not members. Hospitals in the southeast see 160.73 dollars more than hospitals in the northeast, 311.28 dollars less than hospitals in the northcentral, and 362.03 dollars less than hospitals in the mountain region in net income per patient day.

Confirming Section

After testing the hypothesis laid out in the exploring section of this process, I have discovered a statistically significant negative relationship between the thirty day readmission rate for pneumonia patients and net income adjusted for patient days. An increase of 1 percent in the thirty day readmission for pneumonia patients is correlated with a decrease of 61.12 dollars in net income per patient day. Given the median number of patient days for the hospitals in our dataset (approximately 16, 000), the 61 dollars per patient day adds up to a very large change in net income over the course of a year. This negative relationship likely comes from having to care for patients without being paid for that care when the patients are readmitted. The policy that I propose would see CMS offer hospitals that are able to reduce the thirty day readmission rate for pneumonia patients by 5%, a bonus payment of 1%. The bonus payment would be given to the hospital the year following the year that the hospital is able to decrease their readmission rate. To implement this policy, we should measure each hospital's thirty day readmission rate for pneumonia patients from the last five years and use the average of the last five years as our baseline to determine the percent decrease in readmission rate. This policy will allow each hospital to make whatever adjustments they deem necessary for their specific circumstances to lower the readmission rate.

I anticipate that the change in net income due to the change in readmission rate for pneumonia patients will be visible in the first year of implementing the proposed policy. For the experiment, I propose that we randomly select five hospitals from each of the different regions that are specified in the data for our treatment group. The rest of the population will be our control group. We will implement the policy mentioned in the previous paragraph on the treatment group for one year. We will measure the change in income for hospitals by using the average income from the five years before the experiment as a baseline and weighing it against income from the year the policy is implemented. We will be able to look at the difference in the change in income between treatment and control groups. The financial incentive will be applied the year after the experiment takes place to ensure that the financial bonus does not affect the results of the experiment.

If the results of the experiment are positive, the implementation of this policy could lead to huge increases in net income for hospitals in the United States. If the relationship found in the experiment is as strong as the relationship found in the testing section of the process, the increase in net income for a hospital with median patient days would be just over 975,000 dollars in a year. Implementation of the policy would be justified even if the relationship between readmission rate and net income were weaker than what was found in the testing section. The costs associated with implementing this policy will come from

the bonus paid to hospitals in compliance and the labor associated with keeping records on the hospitals to measure compliance. If the results of the experiment are not positive, there may be another quality variable that is driving changes in both pneumonia readmission rates and net income. This may be worth exploring further.

Conclusions

Executive Summary:

After determining several variables of interest, I ran an exploratory analysis of each variable individually and the relationship between every type of variable. I looked for variables that appeared to have strong relationships in order to develop hypotheses to test. I ran regressions on all of my hypothesized relationships to look for statistical significance. I found a statistically significant relationship between the thirty day readmission rate for pneumonia patients and the net income of hospitals within the dataset. I believe that having a high readmission rate causes a decrease in net income due to the requirement to provide care to a patient without being paid for providing that care. I believe that if hospitals are incentivized to lower the readmission rate of pneumonia patients, it will cause an increase in net income.

Conclusion:

I am proposing an experiment to test the relationship between the thirty day readmission rate of pneumonia patients and net income. I believe that if the experiment has positive results, the increase in net income from the recommended policy will greatly outweigh the cost of running the experiment and applying the policy. There is a large statistically significant relationship indicating that a decrease of 5% in pneumonia readmission rate could cause an increase of 305 dollars per patient day. Taking that number by the median number of patient days for a hospital in our data set, the increase in net income would come to approximately 4.89 million dollars. Even if the experiment shows that there is a weaker relationship between the variables by a small margin, implementation of the proposed policy would result in very large increases in net income.